Grade 4 Mathematics

Solved Exercises

Balochistan Textbook Board

Balochistan Textbook Board (BTBB)

Unit 1: Whole Numbers

Unit 1: Whole Numbers. Numbers up to 99,999: place value, expanded form, words and numerals, comparing, ordering, rounding and Roman numerals.

Exercise 1.1

1. Write the place and place value of the coloured digit. (a) 76,423 (coloured digit 6)

(a) The coloured digit is 6. It is in the thousands place, so its place value is 6,000.

(b) 45,328 (coloured digit 3)

(b) The coloured digit is 3. It is in the hundreds place, so its place value is 300.

(c) 82,719 (coloured digit 8)

(c) The coloured digit is 8. It is in the ten-thousands place, so its place value is 80,000.

(d) 91,705 (coloured digit 5)

(d) The coloured digit is 5. It is in the ones place, so its place value is 5.

(e) 64,892 (coloured digit 9)

(e) The coloured digit is 9. It is in the tens place, so its place value is 90.

(f) 23,956 (coloured digit 2)

(f) The coloured digit is 2. It is in the ten-thousands place, so its place value is 20,000.

2. Write 90,230 in expanded form.

90,230 = 90000 + 200 + 30. Each digit is multiplied by its place value and the parts are added.

2. Write 54,109 in expanded form.

54,109 = 50000 + 4000 + 100 + 9. Each digit is multiplied by its place value and the parts are added.

2. Write 82,675 in expanded form.

82,675 = 80000 + 2000 + 600 + 70 + 5. Each digit is multiplied by its place value and the parts are added.

2. Write 61,508 in expanded form.

61,508 = 60000 + 1000 + 500 + 8. Each digit is multiplied by its place value and the parts are added.

2. Write 47,396 in expanded form.

47,396 = 40000 + 7000 + 300 + 90 + 6. Each digit is multiplied by its place value and the parts are added.

3. Write 12,345 in words.

12,345 = Twelve thousand three hundred and forty five.

3. Write 23,678 in words.

23,678 = Twenty three thousand six hundred and seventy eight.

3. Write 34,021 in words.

34,021 = Thirty four thousand and twenty one.

3. Write 50,006 in words.

50,006 = Fifty thousand and six.

3. Write 61,432 in words.

61,432 = Sixty one thousand four hundred and thirty two.

3. Write 72,519 in words.

72,519 = Seventy two thousand five hundred and nineteen.

4. Write in numerals: Twenty one thousand three hundred and forty five.

Twenty one thousand three hundred and forty five = 21,345.

4. Write in numerals: Thirty six thousand seven hundred and twenty one.

Thirty six thousand seven hundred and twenty one = 36,721.

4. Write in numerals: Forty eight thousand and nine.

Forty eight thousand and nine = 48,009.

4. Write in numerals: Fifty five thousand one hundred and seventy.

Fifty five thousand one hundred and seventy = 55,170.

4. Write in numerals: Seventy four thousand five hundred and thirty six.

Seventy four thousand five hundred and thirty six = 74,536.

4. Write in numerals: Eighty nine thousand nine hundred and ninety nine.

Eighty nine thousand nine hundred and ninety nine = 89,999.

Exercise 1.2

1(a). Compare 63,321 and 63,330 using <, > or =.

63,321 < 63,330. Comparing place by place from the left, the ten-thousands digits are equal, so we compare the next places until the digits differ. Note: the book's printed key wrongly gives 13,076 < 10,772; in fact 13,076 > 10,772.

1(b). Compare 74,625 and 74,625 using <, > or =.

74,625 = 74,625. Comparing place by place from the left, both numbers are exactly the same. Note: the book's printed key wrongly gives 13,076 < 10,772; in fact 13,076 > 10,772.

1(c). Compare 13,076 and 10,772 using <, > or =.

13,076 > 10,772. Comparing place by place from the left, the ten-thousands digits are equal, so we compare the next places until the digits differ. Note: the book's printed key wrongly gives 13,076 < 10,772; in fact 13,076 > 10,772.

1(d). Compare 52,473 and 52,462 using <, > or =.

52,473 > 52,462. Comparing place by place from the left, the ten-thousands digits are equal, so we compare the next places until the digits differ. Note: the book's printed key wrongly gives 13,076 < 10,772; in fact 13,076 > 10,772.

1(e). Compare 66,702 and 66,720 using <, > or =.
66,702 < 66,720. Comparing place by place from the left, the ten-thousands digits are equal, so we compare the next places until the digits differ. Note: the book's printed key wrongly gives 13,076 < 10,772; in fact 13,076 > 10,772.

1(f). Compare 52,894 and 23,108 using <, > or =.
52,894 > 23,108. Comparing place by place from the left, the ten-thousands digits are equal, so we compare the next places until the digits differ. Note: the book's printed key wrongly gives 13,076 < 10,772; in fact 13,076 > 10,772.

1(g). Compare 56,410 and 56,410 using <, > or =.
56,410 = 56,410. Comparing place by place from the left, both numbers are exactly the same. Note: the book's printed key wrongly gives 13,076 < 10,772; in fact 13,076 > 10,772.

1(h). Compare 76,425 and 76,425 using <, > or =.
76,425 = 76,425. Comparing place by place from the left, both numbers are exactly the same. Note: the book's printed key wrongly gives 13,076 < 10,772; in fact 13,076 > 10,772.

1(i). Compare 67,588 and 67,529 using <, > or =.

67,588 > 67,529. Comparing place by place from the left, the ten-thousands digits are equal, so we compare the next places until the digits differ. Note: the book's printed key wrongly gives 13,076 < 10,772; in fact 13,076 > 10,772.

1(j). Compare 60,327 and 60,237 using <, > or =.
60,327 > 60,237. Comparing place by place from the left, the ten-thousands digits are equal, so we compare the next places until the digits differ. Note: the book's printed key wrongly gives 13,076 < 10,772; in fact 13,076 > 10,772.

2(a). Write in ascending order: 43,666, 46,326, 47,762.
Ascending order: 43,666, 46,326, 47,762. The smallest number comes first and the largest last.

2(b). Write in ascending order: 35,603, 53,053, 65,053.
Ascending order: 35,603, 53,053, 65,053. The smallest number comes first and the largest last.

2(c). Write in ascending order: 28,202, 28,301, 23,829.
Ascending order: 23,829, 28,202, 28,301. The smallest number comes first and the largest last.

2(d). Write in ascending order: 12,904, 12,508, 11,989, 21,084.
Ascending order: 11,989, 12,508, 12,904, 21,084. The smallest number comes first and the largest last.

2(e). Write in ascending order: 56,488, 32,555, 12,977, 45,366.

Ascending order: 12,977, 32,555, 45,366, 56,488. The smallest number comes first and the largest last.

3(a). Write in descending order: 67,544, 78,439, 76,654.
Descending order: 78,439, 76,654, 67,544. The largest number comes first and the smallest last.

3(b). Write in descending order: 50,098, 50,045, 50,101.
Descending order: 50,101, 50,098, 50,045. The largest number comes first and the smallest last.

3(c). Write in descending order: 34,531, 38,754, 31,208, 38,643.
Descending order: 38,754, 38,643, 34,531, 31,208. The largest number comes first and the smallest last.

3(d). Write in descending order: 56,119, 56,149, 56,139, 56,100.
Descending order: 56,149, 56,139, 56,119, 56,100. The largest number comes first and the smallest last.

3(e). Write in descending order: 34,204, 15,456, 78,957, 54,958.
Descending order: 78,957, 54,958, 34,204, 15,456. The largest number comes first and the smallest last.

Exercise 1.3

1. Tens(a). Round 54 to the nearest tens.
Look at the digit to the right of the tens place. It gives 4, so we round down and replace the digits to the right by zeros. 54 rounds to 50.

1. Tens(b). Round 108 to the nearest tens.

Look at the digit to the right of the tens place. It gives 8, so we round up and replace the digits to the right by zeros. 108 rounds to 110.

1. Tens(c). Round 2,961 to the nearest tens.

Look at the digit to the right of the tens place. It gives 1, so we round down and replace the digits to the right by zeros. 2,961 rounds to 2,960.

1. Tens(d). Round 12,095 to the nearest tens.

Look at the digit to the right of the tens place. It gives 5, so we round up and replace the digits to the right by zeros. 12,095 rounds to 12,100.

2. Hundreds(a). Round 4,596 to the nearest hundreds.

Look at the digit to the right of the hundreds place. It gives 96, so we round up and replace the digits to the right by zeros. 4,596 rounds to 4,600.

2. Hundreds(b). Round 41,427 to the nearest hundreds.

Look at the digit to the right of the hundreds place. It gives 27, so we round down and replace the digits to the right by zeros. 41,427 rounds to 41,400.

2. Hundreds(c). Round 56,781 to the nearest hundreds.

Look at the digit to the right of the hundreds place. It gives 81, so

we round up and replace the digits to the right by zeros. 56,781 rounds to 56,800.

2. Hundreds(d). Round 17,461 to the nearest hundreds.
Look at the digit to the right of the hundreds place. It gives 61, so we round up and replace the digits to the right by zeros. 17,461 rounds to 17,500.

3. Thousands(a). Round 3,506 to the nearest thousands.
Look at the digit to the right of the thousands place. It gives 506, so we round up and replace the digits to the right by zeros. 3,506 rounds to 4,000.

3. Thousands(b). Round 12,345 to the nearest thousands.
Look at the digit to the right of the thousands place. It gives 345, so we round down and replace the digits to the right by zeros. 12,345 rounds to 12,000.

3. Thousands(c). Round 19,008 to the nearest thousands.
Look at the digit to the right of the thousands place. It gives 8, so we round down and replace the digits to the right by zeros. 19,008 rounds to 19,000.

3. Thousands(d). Round 42,750 to the nearest thousands.
Look at the digit to the right of the thousands place. It gives 750, so we round up and replace the digits to the right by zeros. 42,750 rounds to 43,000.

4. Ten-thousands(a). Round 12,345 to the nearest ten-thousands.
Look at the digit to the right of the ten-thousands place. It gives 2345, so we round down and replace the digits to the right by zeros. 12,345 rounds to 10,000.

4. Ten-thousands(b). Round 76,498 to the nearest ten-thousands.
Look at the digit to the right of the ten-thousands place. It gives 6498, so we round up and replace the digits to the right by zeros. 76,498 rounds to 80,000.

4. Ten-thousands(c). Round 45,009 to the nearest ten-thousands.
Look at the digit to the right of the ten-thousands place. It gives 5009, so we round up and replace the digits to the right by zeros. 45,009 rounds to 50,000.

4. Ten-thousands(d). Round 65,768 to the nearest ten-thousands.
Look at the digit to the right of the ten-thousands place. It gives 5768, so we round up and replace the digits to the right by zeros. 65,768 rounds to 70,000.

Exercise 1.4

1(a). Express XXXV in Arabic numerals.
XXXV = 35. Break the Roman numeral into its symbols and add or subtract by the rules.

1(b). Express XL in Arabic numerals.
XL = 40. Break the Roman numeral into its symbols and add or subtract by the rules.

1(c). Express XLIV in Arabic numerals.

XLIV = 44. Break the Roman numeral into its symbols and add or subtract by the rules.

1(d). Express XXX in Arabic numerals.

XXX = 30. Break the Roman numeral into its symbols and add or subtract by the rules.

1(e). Express LXXXIV in Arabic numerals.

LXXXIV = 84. Break the Roman numeral into its symbols and add or subtract by the rules.

1(f). Express XLI in Arabic numerals.

XLI = 41. Break the Roman numeral into its symbols and add or subtract by the rules.

1(g). Express L in Arabic numerals.

L = 50. Break the Roman numeral into its symbols and add or subtract by the rules.

1(h). Express XXV in Arabic numerals.

XXV = 25. Break the Roman numeral into its symbols and add or subtract by the rules.

1(i). Express XLVII in Arabic numerals.

XLVII = 47. Break the Roman numeral into its symbols and add or subtract by the rules.

1(j). Express LXVI in Arabic numerals.

LXVI = 66. Break the Roman numeral into its symbols and add or subtract by the rules.

2(a). Write 22 in Roman numerals.

22 = XXII. Build the number from tens and ones using the Roman symbols.

2(b). Write 17 in Roman numerals.

17 = XVII. Build the number from tens and ones using the Roman symbols.

2(c). Write 64 in Roman numerals.

64 = LXIV. Build the number from tens and ones using the Roman symbols.

2(d). Write 55 in Roman numerals.

55 = LV. Build the number from tens and ones using the Roman symbols.

2(e). Write 89 in Roman numerals.

89 = LXXXIX. Build the number from tens and ones using the Roman symbols.

2(f). Write 41 in Roman numerals.

41 = XLI. Build the number from tens and ones using the Roman symbols.

2(g). Write 76 in Roman numerals.

76 = LXXVI. Build the number from tens and ones using the Roman symbols.

2(h). Write 33 in Roman numerals.

33 = XXXIII. Build the number from tens and ones using the Roman symbols.

2(i). Write 20 in Roman numerals.

20 = XX. Build the number from tens and ones using the Roman symbols.

2(j). Write 92 in Roman numerals.

92 = XCII. Build the number from tens and ones using the Roman symbols.

Review Exercise 1

1. Place and place value of the underlined digit. (a) 12,360 (1 underlined)

(a) The underlined digit is 1, in the ten-thousands place. Place value = 10,000.

(b) 61,474 (last 4 underlined)

(b) The underlined digit is 4, in the ones place. Place value = 4.

(c) 56,324 (5 underlined)

(c) The underlined digit is 5, in the ten-thousands place. Place value

= 50,000.

(d) 47,490 (9 underlined)

(d) The underlined digit is 9, in the tens place. Place value = 90.

(e) 98,194 (1 underlined)

(e) The underlined digit is 1, in the hundreds place. Place value = 100.

2. Write 83,764 in words.

83,764 = Eighty three thousand seven hundred and sixty four.

2. Write 94,208 in words.

94,208 = Ninety four thousand two hundred and eight.

2. Write 98,963 in words.

98,963 = Ninety eight thousand nine hundred and sixty three.

3. Write in numerals: Ninety one thousand and fifteen.

Ninety one thousand and fifteen = 91,015.

3. Write in numerals: Ten thousand five hundred and forty two.

Ten thousand five hundred and forty two = 10,542.

3. Write in numerals: Ninety nine thousand eight hundred and seventy.

Ninety nine thousand eight hundred and seventy = 99,870.

4(a). Ascending and descending: 83,401; 97,035; 12,337.

Ascending order: 12,337, 83,401, 97,035. Descending order: 97,035, 83,401, 12,337.

4(b). Ascending and descending: 18,017; 18,221; 13,411.

Ascending order: 13,411, 18,017, 18,221. Descending order: 18,221, 18,017, 13,411.

5. Compare: (a) 84,325 and 93,417; (b) 68,709 and 43,216; (c) 56,708 and 32,156; (d) 74,932 and 74,932.

(a) 84,325 < 93,417, because 8 ten-thousands is less than 9 ten-thousands. (b) 68,709 > 43,216, because 6 ten-thousands is greater than 4 ten-thousands. (c) 56,708 > 32,156, because 5 ten-thousands is greater than 3 ten-thousands. (d) 74,932 = 74,932, because both numbers are exactly the same.

6(i). Round to the nearest ten: 77, 203, 3,841, 12,467.

(a) 77 rounds to 80. (b) 203 rounds to 200. (c) 3,841 rounds to 3,840. (d) 12,467 rounds to 12,470. Look at the ones digit: 5 or more rounds up, less than 5 rounds down.

6(ii). Round to the nearest hundred: 3,475, 52,538, 67,892, 64,573.

(a) 3,475 rounds to 3,500. (b) 52,538 rounds to 52,500. (c) 67,892 rounds to 67,900. (d) 64,573 rounds to 64,600. Look at the tens digit: 5 or more rounds up, less than 5 rounds down.

6(iii). Round to the nearest thousand: 4,617, 23,456, 38,113, 53,267.

(a) 4,617 rounds to 5,000. (b) 23,456 rounds to 23,000. (c) 38,113 rounds to 38,000. (d) 53,267 rounds to 53,000. Look at the hundreds digit: 5 or more rounds up, less than 5 rounds down.

6(iv). Round to the nearest ten-thousand: 12,546, 65,329, 50,721, 19,005.

(a) 12,546 rounds to 10,000. (b) 65,329 rounds to 70,000. (c) 50,721 rounds to 50,000. (d) 19,005 rounds to 20,000. Look at the thousands digit: 5 or more rounds up, less than 5 rounds down.

Unit 2: Addition and Subtraction

Unit 2: Addition and Subtraction. Add and subtract 5-digit numbers with and without regrouping, estimate by rounding, and solve word problems.

Exercise 2.1 - Add the following numbers

1(a). Add 68,576 + 3.

Add the ones: 6 + 3 = 9. Write 9.

Add the tens: 7 + 0 = 7. Write 7.

Add the hundreds: 5 + 0 = 5. Write 5.

Add the thousands: 8 + 0 = 8. Write 8.

Add the ten-thousands: 6 + 0 = 6. Write 6.

Answer: 68,576 + 3 = 68,579.

1(b). Add 77,904 + 95.

Add the ones: 4 + 5 = 9. Write 9.

Add the tens: 0 + 9 = 9. Write 9.

Add the hundreds: 9 + 0 = 9. Write 9.

Add the thousands: 7 + 0 = 7. Write 7.

Add the ten-thousands: 7 + 0 = 7. Write 7.

Answer: 77,904 + 95 = 77,999.

1(c). Add 88,337 + 550.

Add the ones: 7 + 0 = 7. Write 7.

Add the tens: 3 + 5 = 8. Write 8.

Add the hundreds: 3 + 5 = 8. Write 8.

Add the thousands: 8 + 0 = 8. Write 8.

Add the ten-thousands: 8 + 0 = 8. Write 8.

Answer: 88,337 + 550 = 88,887.

1(d). Add 54,831 + 5,067.

Add the ones: 1 + 7 = 8. Write 8.

Add the tens: 3 + 6 = 9. Write 9.

Add the hundreds: 8 + 0 = 8. Write 8.

Add the thousands: 4 + 5 = 9. Write 9.

Add the ten-thousands: 5 + 0 = 5. Write 5.

Answer: 54,831 + 5,067 = 59,898.

1(e). Add 54,539 + 43,210.

Add the ones: 9 + 0 = 9. Write 9.

Add the tens: 3 + 1 = 4. Write 4.

Add the hundreds: 5 + 2 = 7. Write 7.

Add the thousands: 4 + 3 = 7. Write 7.

Add the ten-thousands: 5 + 4 = 9. Write 9.

Answer: 54,539 + 43,210 = 97,749.

1(f). Add 75,346 + 10,152.

Add the ones: 6 + 2 = 8. Write 8.

Add the tens: 4 + 5 = 9. Write 9.

Add the hundreds: 3 + 1 = 4. Write 4.

Add the thousands: 5 + 0 = 5. Write 5.

Add the ten-thousands: 7 + 1 = 8. Write 8.

Answer: 75,346 + 10,152 = 85,498.

Exercise 2.2

1(a). Add 75,486 + 6.

Add the ones: 6 + 6 = 12. Write 2 and carry 1.

Add the tens: 8 + 0 + 1 (carried) = 9. Write 9.

Add the hundreds: 4 + 0 = 4. Write 4.

Add the thousands: 5 + 0 = 5. Write 5.

Add the ten-thousands: 7 + 0 = 7. Write 7.

Answer: 75,486 + 6 = 75,492.

1(b). Add 53,585 + 87.

Add the ones: 5 + 7 = 12. Write 2 and carry 1.

Add the tens: 8 + 8 + 1 (carried) = 17. Write 7 and carry 1.

Add the hundreds: 5 + 0 + 1 (carried) = 6. Write 6.

Add the thousands: 3 + 0 = 3. Write 3.

Add the ten-thousands: 5 + 0 = 5. Write 5.

Answer: 53,585 + 87 = 53,672.

1(c). Add 36,593 + 521.

Add the ones: 3 + 1 = 4. Write 4.

Add the tens: 9 + 2 = 11. Write 1 and carry 1.

Add the hundreds: 5 + 5 + 1 (carried) = 11. Write 1 and carry 1.

Add the thousands: 6 + 0 + 1 (carried) = 7. Write 7.

Add the ten-thousands: 3 + 0 = 3. Write 3.

Answer: 36,593 + 521 = 37,114.

1(d). Add 52,793 + 8,857.

Add the ones: 3 + 7 = 10. Write 0 and carry 1.

Add the tens: 9 + 5 + 1 (carried) = 15. Write 5 and carry 1.

Add the hundreds: 7 + 8 + 1 (carried) = 16. Write 6 and carry 1.

Add the thousands: 2 + 8 + 1 (carried) = 11. Write 1 and carry 1.

Add the ten-thousands: 5 + 0 + 1 (carried) = 6. Write 6.

Answer: 52,793 + 8,857 = 61,650.

1(e). Add 43,586 + 26,935.

Add the ones: 6 + 5 = 11. Write 1 and carry 1.

Add the tens: 8 + 3 + 1 (carried) = 12. Write 2 and carry 1.

Add the hundreds: 5 + 9 + 1 (carried) = 15. Write 5 and carry 1.

Add the thousands: 3 + 6 + 1 (carried) = 10. Write 0 and carry 1.

Add the ten-thousands: 4 + 2 + 1 (carried) = 7. Write 7.

Answer: 43,586 + 26,935 = 70,521.

1(f). Add 34,507 + 26,893.

Add the ones: 7 + 3 = 10. Write 0 and carry 1.

Add the tens: 0 + 9 + 1 (carried) = 10. Write 0 and carry 1.

Add the hundreds: 5 + 8 + 1 (carried) = 14. Write 4 and carry 1.

Add the thousands: 4 + 6 + 1 (carried) = 11. Write 1 and carry 1.

Add the ten-thousands: 3 + 2 + 1 (carried) = 6. Write 6.

Answer: 34,507 + 26,893 = 61,400.

2. Nauman bought a washing machine for Rs. 68,950 and an oven for Rs. 21,550. How much money did he spend altogether?

Money spent = price of washing machine + price of oven.

Add the ones: 0 + 0 = 0. Write 0.

Add the tens: 5 + 5 = 10. Write 0 and carry 1.

Add the hundreds: 9 + 5 + 1 (carried) = 15. Write 5 and carry 1.

Add the thousands: 8 + 1 + 1 (carried) = 10. Write 0 and carry 1.

Add the ten-thousands: 6 + 2 + 1 (carried) = 9. Write 9.

Answer: Rs. 68,950 + Rs. 21,550 = Rs. 90,500. Nauman spent Rs. 90,500 altogether.

3. A factory manufactured 57,437 electric cars in one month and 41,745 in the second month. How many cars in both months?

Total cars = cars in first month + cars in second month.

Add the ones: 7 + 5 = 12. Write 2 and carry 1.

Add the tens: 3 + 4 + 1 (carried) = 8. Write 8.

Add the hundreds: 4 + 7 = 11. Write 1 and carry 1.

Add the thousands: 7 + 1 + 1 (carried) = 9. Write 9.

Add the ten-thousands: 5 + 4 = 9. Write 9.

Answer: 57,437 + 41,745 = 99,182. The factory manufactured 99,182 cars in both months.

4. There are Rs. 15,785 in Hamza's father's account. He credits Rs. 4,898. How much is in the account now?

Total amount = old balance + credited amount.

Add the ones: 5 + 8 = 13. Write 3 and carry 1.

Add the tens: 8 + 9 + 1 (carried) = 18. Write 8 and carry 1.

Add the hundreds: 7 + 8 + 1 (carried) = 16. Write 6 and carry 1.

Add the thousands: 5 + 4 + 1 (carried) = 10. Write 0 and carry 1.

Add the ten-thousands: 1 + 0 + 1 (carried) = 2. Write 2.

Answer: Rs. 15,785 + Rs. 4,898 = Rs. 20,683. Now he has Rs. 20,683.

Exercise 2.3 - Subtract the following 5-digit numbers

1(a). Subtract 5 from 34,687.

Subtract the ones: 7 - 5 = 2. Write 2.

Subtract the tens: 8 - 0 = 8. Write 8.

Subtract the hundreds: 6 - 0 = 6. Write 6.

Subtract the thousands: 4 - 0 = 4. Write 4.

Subtract the ten-thousands: 3 - 0 = 3. Write 3.

Answer: 34,687 - 5 = 34,682.

1(b). Subtract 54 from 78,965.

Subtract the ones: 5 - 4 = 1. Write 1.

Subtract the tens: 6 - 5 = 1. Write 1.

Subtract the hundreds: 9 - 0 = 9. Write 9.

Subtract the thousands: 8 - 0 = 8. Write 8.

Subtract the ten-thousands: 7 - 0 = 7. Write 7.

Answer: 78,965 - 54 = 78,911.

1(c). Subtract 724 from 50,947.

Subtract the ones: 7 - 4 = 3. Write 3.

Subtract the tens: 4 - 2 = 2. Write 2.

Subtract the hundreds: 9 - 7 = 2. Write 2.

Subtract the thousands: 0 - 0 = 0. Write 0.

Subtract the ten-thousands: 5 - 0 = 5. Write 5.

Answer: 50,947 - 724 = 50,223.

1(d). Subtract 5,174 from 97,486.

Subtract the ones: 6 - 4 = 2. Write 2.

Subtract the tens: 8 - 7 = 1. Write 1.

Subtract the hundreds: 4 - 1 = 3. Write 3.

Subtract the thousands: 7 - 5 = 2. Write 2.

Subtract the ten-thousands: 9 - 0 = 9. Write 9.

Answer: 97,486 - 5,174 = 92,312.

1(e). Subtract 18,276 from 79,378.

Subtract the ones: 8 - 6 = 2. Write 2.

Subtract the tens: 7 - 7 = 0. Write 0.

Subtract the hundreds: 3 - 2 = 1. Write 1.

Subtract the thousands: 9 - 8 = 1. Write 1.

Subtract the ten-thousands: 7 - 1 = 6. Write 6.

Answer: 79,378 - 18,276 = 61,102.

1(f). Subtract 32,135 from 48,446.

Subtract the ones: 6 - 5 = 1. Write 1.

Subtract the tens: 4 - 3 = 1. Write 1.

Subtract the hundreds: 4 - 1 = 3. Write 3.

Subtract the thousands: 8 - 2 = 6. Write 6.

Subtract the ten-thousands: 4 - 3 = 1. Write 1.

Answer: 48,446 - 32,135 = 16,311.

Exercise 2.4 - Subtract the following 5-digit numbers

1(a). Subtract 8 from 65,730.

Subtract the ones: 0 is smaller than 8, so borrow 1 from the tens.

10 - 8 = 2. Write 2.

Subtract the tens: 2 - 0 = 2. Write 2.

Subtract the hundreds: 7 - 0 = 7. Write 7.

Subtract the thousands: 5 - 0 = 5. Write 5.

Subtract the ten-thousands: 6 - 0 = 6. Write 6.

Answer: 65,730 - 8 = 65,722.

1(b). Subtract 97 from 77,484.

Subtract the ones: 4 is smaller than 7, so borrow 1 from the tens. 14 - 7 = 7. Write 7.

Subtract the tens: 7 is smaller than 9, so borrow 1 from the hundreds. 17 - 9 = 8. Write 8.

Subtract the hundreds: 3 - 0 = 3. Write 3.

Subtract the thousands: 7 - 0 = 7. Write 7.

Subtract the ten-thousands: 7 - 0 = 7. Write 7.

Answer: 77,484 - 97 = 77,387.

1(c). Subtract 587 from 84,375.

Subtract the ones: 5 is smaller than 7, so borrow 1 from the tens. 15 - 7 = 8. Write 8.

Subtract the tens: 6 is smaller than 8, so borrow 1 from the hundreds. 16 - 8 = 8. Write 8.

Subtract the hundreds: 2 is smaller than 5, so borrow 1 from the thousands. 12 - 5 = 7. Write 7.

Subtract the thousands: 3 - 0 = 3. Write 3.

Subtract the ten-thousands: 8 - 0 = 8. Write 8.

Answer: 84,375 - 587 = 83,788.

1(d). Subtract 3,797 from 93,274.

Subtract the ones: 4 is smaller than 7, so borrow 1 from the tens. 14 - 7 = 7. Write 7.

Subtract the tens: 6 is smaller than 9, so borrow 1 from the hundreds. 16 - 9 = 7. Write 7.

Subtract the hundreds: 1 is smaller than 7, so borrow 1 from the thousands. 11 - 7 = 4. Write 4.

Subtract the thousands: 2 is smaller than 3, so borrow 1 from the ten-thousands. 12 - 3 = 9. Write 9.

Subtract the ten-thousands: 8 - 0 = 8. Write 8.

Answer: 93,274 - 3,797 = 89,477.

1(e). Subtract 5,697 from 63,472.

Subtract the ones: 2 is smaller than 7, so borrow 1 from the tens. 12 - 7 = 5. Write 5.

Subtract the tens: 6 is smaller than 9, so borrow 1 from the hundreds. 16 - 9 = 7. Write 7.

Subtract the hundreds: 3 is smaller than 6, so borrow 1 from the thousands. 13 - 6 = 7. Write 7.

Subtract the thousands: 2 is smaller than 5, so borrow 1 from the ten-thousands. 12 - 5 = 7. Write 7.

Subtract the ten-thousands: 5 - 0 = 5. Write 5.

Answer: 63,472 - 5,697 = 57,775.

1(f). Subtract 43,568 from 71,426.

Subtract the ones: 6 is smaller than 8, so borrow 1 from the tens. 16 - 8 = 8. Write 8.

Subtract the tens: 1 is smaller than 6, so borrow 1 from the hundreds. 11 - 6 = 5. Write 5.

Subtract the hundreds: 3 is smaller than 5, so borrow 1 from the thousands. 13 - 5 = 8. Write 8.

Subtract the thousands: 0 is smaller than 3, so borrow 1 from the ten-thousands. 10 - 3 = 7. Write 7.

Subtract the ten-thousands: 6 - 4 = 2. Write 2.

Answer: 71,426 - 43,568 = 27,858.

2. Aliya had Rs. 65,785. She bought items for Rs. 54,894. What amount does she have left?

Amount left = money Aliya had - money spent.

Subtract the ones: 5 - 4 = 1. Write 1.

Subtract the tens: 8 is smaller than 9, so borrow 1 from the hundreds. 18 - 9 = 9. Write 9.

Subtract the hundreds: 6 is smaller than 8, so borrow 1 from the thousands. 16 - 8 = 8. Write 8.

Subtract the thousands: 4 - 4 = 0. Write 0.

Subtract the ten-thousands: 6 - 5 = 1. Write 1.

Answer: Rs. 65,785 - Rs. 54,894 = Rs. 10,891. Aliya has Rs. 10,891 left.

3. A farmer grew 86,731 kg wheat in 2019 and 64,347 kg in 2020. How many kg less did he grow in 2020?

Less wheat = 2019 crop - 2020 crop.

Subtract the ones: 1 is smaller than 7, so borrow 1 from the tens. 11 - 7 = 4. Write 4.

Subtract the tens: 2 is smaller than 4, so borrow 1 from the hundreds. 12 - 4 = 8. Write 8.

Subtract the hundreds: 6 - 3 = 3. Write 3.

Subtract the thousands: 6 - 4 = 2. Write 2.

Subtract the ten-thousands: 8 - 6 = 2. Write 2.

Answer: 86,731 - 64,347 = 22,384. He grew 22,384 kg less wheat in 2020.

Review Exercise 2

1(a). Add 43,253 + 33,155.

Add the ones: 3 + 5 = 8. Write 8.

Add the tens: 5 + 5 = 10. Write 0 and carry 1.

Add the hundreds: 2 + 1 + 1 (carried) = 4. Write 4.

Add the thousands: 3 + 3 = 6. Write 6.

Add the ten-thousands: 4 + 3 = 7. Write 7.

Answer: 43,253 + 33,155 = 76,408.

1(b). Add 54,963 + 32,785.

Add the ones: 3 + 5 = 8. Write 8.

Add the tens: 6 + 8 = 14. Write 4 and carry 1.

Add the hundreds: 9 + 7 + 1 (carried) = 17. Write 7 and carry 1.

Add the thousands: 4 + 2 + 1 (carried) = 7. Write 7.

Add the ten-thousands: 5 + 3 = 8. Write 8.

Answer: 54,963 + 32,785 = 87,748.

1(c). Add 56,781 + 14,950.

Add the ones: 1 + 0 = 1. Write 1.

Add the tens: 8 + 5 = 13. Write 3 and carry 1.

Add the hundreds: 7 + 9 + 1 (carried) = 17. Write 7 and carry 1.

Add the thousands: 6 + 4 + 1 (carried) = 11. Write 1 and carry 1.

Add the ten-thousands: 5 + 1 + 1 (carried) = 7. Write 7.

Answer: 56,781 + 14,950 = 71,731.

1(d). Add 63,059 + 28,912.

Add the ones: 9 + 2 = 11. Write 1 and carry 1.

Add the tens: 5 + 1 + 1 (carried) = 7. Write 7.

Add the hundreds: 0 + 9 = 9. Write 9.

Add the thousands: 3 + 8 = 11. Write 1 and carry 1.

Add the ten-thousands: 6 + 2 + 1 (carried) = 9. Write 9.

Answer: 63,059 + 28,912 = 91,971.

2(a). Subtract 29,478 from 58,341.

Subtract the ones: 1 is smaller than 8, so borrow 1 from the tens. 11 - 8 = 3. Write 3.

Subtract the tens: 3 is smaller than 7, so borrow 1 from the hundreds. 13 - 7 = 6. Write 6.

Subtract the hundreds: 2 is smaller than 4, so borrow 1 from the thousands. 12 - 4 = 8. Write 8.

Subtract the thousands: 7 is smaller than 9, so borrow 1 from the ten-thousands. 17 - 9 = 8. Write 8.

Subtract the ten-thousands: 4 - 2 = 2. Write 2.

Answer: 58,341 - 29,478 = 28,863.

2(b). Subtract 31,757 from 83,268.

Subtract the ones: 8 - 7 = 1. Write 1.

Subtract the tens: 6 - 5 = 1. Write 1.

Subtract the hundreds: 2 is smaller than 7, so borrow 1 from the thousands. 12 - 7 = 5. Write 5.

Subtract the thousands: 2 - 1 = 1. Write 1.

Subtract the ten-thousands: 8 - 3 = 5. Write 5.

Answer: 83,268 - 31,757 = 51,511.

2(c). Subtract 11,884 from 27,925.

Subtract the ones: 5 - 4 = 1. Write 1.

Subtract the tens: 2 is smaller than 8, so borrow 1 from the hundreds. 12 - 8 = 4. Write 4.

Subtract the hundreds: 8 - 8 = 0. Write 0.

Subtract the thousands: 7 - 1 = 6. Write 6.

Subtract the ten-thousands: 2 - 1 = 1. Write 1.

Answer: 27,925 - 11,884 = 16,041.

2(d). Subtract 31,261 from 85,429.

Subtract the ones: 9 - 1 = 8. Write 8.

Subtract the tens: 2 is smaller than 6, so borrow 1 from the

hundreds. 12 - 6 = 6. Write 6.

Subtract the hundreds: 3 - 2 = 1. Write 1.

Subtract the thousands: 5 - 1 = 4. Write 4.

Subtract the ten-thousands: 8 - 3 = 5. Write 5.

Answer: 85,429 - 31,261 = 54,168.

3(i). Estimate the sum of 54,963 + 32,785 by rounding.

Exact sum = 87,748. Estimates: nearest tens: 54,960 + 32,790 = 87,750; nearest hundreds: 55,000 + 32,800 = 87,800; nearest thousands: 55,000 + 33,000 = 88,000; nearest ten-thousands: 50,000 + 30,000 = 80,000.

3(ii). Estimate the sum of 43,253 + 33,155 by rounding.

Exact sum = 76,408. Estimates: nearest tens: 43,250 + 33,160 = 76,410; nearest hundreds: 43,300 + 33,200 = 76,500; nearest thousands: 43,000 + 33,000 = 76,000; nearest ten-thousands: 40,000 + 30,000 = 70,000.

3(iii). Estimate the difference of 27,925 - 11,884 by rounding.

Exact difference = 16,041. Estimates: nearest tens: 27,930 - 11,880 = 16,050; nearest hundreds: 27,900 - 11,900 = 16,000; nearest thousands: 28,000 - 12,000 = 16,000; nearest ten-thousands: 30,000 - 10,000 = 20,000.

3(iv). Estimate the difference of 84,531 - 57,625 by rounding.

Exact difference = 26,906. Estimates: nearest tens: 84,530 - 57,630 = 26,900; nearest hundreds: 84,500 - 57,600 = 26,900; nearest thousands: 85,000 - 58,000 = 27,000; nearest ten-thousands: 80,000 - 60,000 = 20,000.

4(i). Mentally: 6,623 + 6,729 = ?

6,623 + 6,729 = 13,352. Add or subtract place by place from the left, keeping the running total in mind.

4(ii). Mentally: 8,506 - 3,499 = ?

8,506 - 3,499 = 5,007. Add or subtract place by place from the left, keeping the running total in mind.

4(iii). Mentally: 6,729 + 3,322 = ?

6,729 + 3,322 = 10,051. Add or subtract place by place from the left, keeping the running total in mind.

4(iv). Mentally: 9,664 - 3,457 = ?

9,664 - 3,457 = 6,207. Add or subtract place by place from the left, keeping the running total in mind.

4(v). Mentally: 1,152 + 1,829 = ?

1,152 + 1,829 = 2,981. Add or subtract place by place from the left, keeping the running total in mind.

4(vi). Mentally: 4,888 - 2,999 = ?

4,888 - 2,999 = 1,889. Add or subtract place by place from the left,

keeping the running total in mind.

4(vii). Mentally: 2,240 + 3,340 = ?

2,240 + 3,340 = 5,580. Add or subtract place by place from the left, keeping the running total in mind.

4(viii). Mentally: 7,800 - 5,214 = ?

7,800 - 5,214 = 2,586. Add or subtract place by place from the left, keeping the running total in mind.

5. A factory made 32,645 shopping bags on Monday and 45,270 more on Tuesday. How many bags were made altogether?

Bags on Tuesday = 32,645 + 45,270 = 77,915, because Tuesday's production is 45,270 more than Monday's.

Total bags = Monday's bags + Tuesday's bags.

Add the ones: 5 + 5 = 10. Write 0 and carry 1.

Add the tens: 4 + 1 + 1 (carried) = 6. Write 6.

Add the hundreds: 6 + 9 = 15. Write 5 and carry 1.

Add the thousands: 2 + 7 + 1 (carried) = 10. Write 0 and carry 1.

Add the ten-thousands: 3 + 7 + 1 (carried) = 11. Write 1 and carry 1.

Carry 1 to the next place.

Answer: 32,645 + 77,915 = 110,560. Altogether 110,560 shopping bags were made.

6. A school needs Rs. 45,550. Students collected Rs. 16,027. How much more money must be collected?
Money still needed = total needed - money collected.

Subtract the ones: 0 is smaller than 7, so borrow 1 from the tens. 10 - 7 = 3. Write 3.

Subtract the tens: 4 - 2 = 2. Write 2.

Subtract the hundreds: 5 - 0 = 5. Write 5.

Subtract the thousands: 5 is smaller than 6, so borrow 1 from the ten-thousands. 15 - 6 = 9. Write 9.

Subtract the ten-thousands: 3 - 1 = 2. Write 2.

Answer: Rs. 45,550 - Rs. 16,027 = Rs. 29,523. Rs. 29,523 more must be collected.

7. A school library has 25,275 books in Urdu and 13,150 books in English. What is the total number of books?
Total books = Urdu books + English books.

Add the ones: 5 + 0 = 5. Write 5.

Add the tens: 7 + 5 = 12. Write 2 and carry 1.

Add the hundreds: 2 + 1 + 1 (carried) = 4. Write 4.

Add the thousands: 5 + 3 = 8. Write 8.

Add the ten-thousands: 2 + 1 = 3. Write 3.

Answer: 25,275 + 13,150 = 38,425. The library has 38,425 books in total.

8. Zahid earns Rs. 39,523 in a month. His expenses are Rs. 25,279. How much does he save in a month?

Savings = earnings - expenses.

Subtract the ones: 3 is smaller than 9, so borrow 1 from the tens. 13 - 9 = 4. Write 4.

Subtract the tens: 1 is smaller than 7, so borrow 1 from the hundreds. 11 - 7 = 4. Write 4.

Subtract the hundreds: 4 - 2 = 2. Write 2.

Subtract the thousands: 9 - 5 = 4. Write 4.

Subtract the ten-thousands: 3 - 2 = 1. Write 1.

Answer: Rs. 39,523 - Rs. 25,279 = Rs. 14,244. Zahid saves Rs. 14,244 in a month.

Unit 3: Multiplication and Division

Unit 3: Multiplication and Division. Multiply 4-digit numbers by 1-digit and 2-digit numbers, divide 4-digit numbers by 1-digit and 2-digit numbers, and solve word problems.

Exercise 3.1

1(i). Multiply 7,462 x 6.

Multiply 7,462 by 6.

7,462 x 6 = 44,772 (multiply each digit by 6, carrying where needed).

Answer: 7,462 x 6 = 44,772.

1(ii). Multiply 3,521 x 23.

Multiply 3,521 by 23.

Break 23 into 20 and 3.

3,521 x 3 = 10,563.

3,521 x 20 = 70,420.

Add the partial products: 10,563 + 70,420 = 80,983.

Answer: 3,521 x 23 = 80,983.

1(iii). Multiply 1,234 x 47.

Multiply 1,234 by 47.

Break 47 into 40 and 7.

1,234 x 7 = 8,638.

1,234 x 40 = 49,360.

Add the partial products: 8,638 + 49,360 = 57,998.

Answer: 1,234 x 47 = 57,998.

1(iv). Multiply 5,736 x 17.

Multiply 5,736 by 17.

Break 17 into 10 and 7.

5,736 x 7 = 40,152.

5,736 x 10 = 57,360.

Add the partial products: 40,152 + 57,360 = 97,512.

Answer: 5,736 x 17 = 97,512.

1(v). Multiply 1,412 x 40.

Multiply 1,412 by 40.

Break 40 into 40 and 0.

1,412 x 0 = 0.

1,412 x 40 = 56,480.

Add the partial products: 0 + 56,480 = 56,480.

Answer: 1,412 x 40 = 56,480.

1(vi). Multiply 7,250 x 3.

Multiply 7,250 by 3.

7,250 x 3 = 21,750 (multiply each digit by 3, carrying where needed).

Answer: 7,250 x 3 = 21,750.

2. The cost of a toy is Rs. 4,410. Find the cost of 7 such toys.

Multiply 4,410 by 7.

4,410 x 7 = 30,870 (multiply each digit by 7, carrying where needed).

Answer: Rs. 4,410 x 7 = Rs. 30,870. Seven toys cost Rs. 30,870.

3. There are 1,215 beads in a tasbeeh. How many beads are there in 54 such tasbeehs?

Multiply 1,215 by 54.

Break 54 into 50 and 4.

1,215 x 4 = 4,860.

1,215 x 50 = 60,750.

Add the partial products: 4,860 + 60,750 = 65,610.

Answer: 1,215 x 54 = 65,610. There are 65,610 beads in 54 tasbeehs.

Exercise 3.2

1(i). Divide 1,845 by 15.

Divide 1,845 by 15 by long division, starting from the highest place.

15 goes into 1,845 123 times (15 x 123 = 1,845).

Subtract: 1,845 - 1,845 = 0. Remainder = 0.

Answer: 1,845 / 15 = 123 exactly.

1(ii). Divide 4,594 by 33.

Divide 4,594 by 33 by long division, starting from the highest place.

33 goes into 4,594 139 times (33 x 139 = 4,587).

Subtract: 4,594 - 4,587 = 7. Remainder = 7.

Answer: 4,594 / 33 = 139, remainder 7.

1(iii). Divide 6,918 by 3.

Divide 6,918 by 3 by long division, starting from the highest place.

3 goes into 6,918 2,306 times (3 x 2,306 = 6,918).

Subtract: 6,918 - 6,918 = 0. Remainder = 0.

Answer: 6,918 / 3 = 2,306 exactly.

1(iv). Divide 1,944 by 27.

Divide 1,944 by 27 by long division, starting from the highest place.

27 goes into 1,944 72 times (27 x 72 = 1,944).

Subtract: 1,944 - 1,944 = 0. Remainder = 0.

Answer: 1,944 / 27 = 72 exactly.

1(v). Divide 8,972 by 81.

Divide 8,972 by 81 by long division, starting from the highest place.

81 goes into 8,972 110 times (81 x 110 = 8,910).

Subtract: 8,972 - 8,910 = 62. Remainder = 62.

Answer: 8,972 / 81 = 110, remainder 62.

1(vi). Divide 6,132 by 7.

Divide 6,132 by 7 by long division, starting from the highest place.

7 goes into 6,132 876 times (7 x 876 = 6,132).

Subtract: 6,132 - 6,132 = 0. Remainder = 0.

Answer: 6,132 / 7 = 876 exactly.

2. 8,240 gift boxes are to be distributed among 20 children. How many boxes did each child get?

Divide 8,240 by 20 by long division, starting from the highest place.

20 goes into 8,240 412 times (20 x 412 = 8,240).

Subtract: 8,240 - 8,240 = 0. Remainder = 0.

Answer: 8,240 / 20 = 412. Each child got 412 gift boxes.

3. The cost of 21 toy cars is Rs. 4,956. Find the cost of one toy car.

Divide 4,956 by 21 by long division, starting from the highest place.

21 goes into 4,956 236 times (21 x 236 = 4,956).

Subtract: 4,956 - 4,956 = 0. Remainder = 0.

Answer: Rs. 4,956 / 21 = Rs. 236. One toy car costs Rs. 236.

Review Exercise 3

1(i). Multiply mentally: 1,234 x 3.

Multiply 1,234 by 3.

1,234 x 3 = 3,702 (multiply each digit by 3, carrying where needed).

Answer: 1,234 x 3 = 3,702.

1(ii). Multiply mentally: 3,651 x 7.

Multiply 3,651 by 7.

3,651 x 7 = 25,557 (multiply each digit by 7, carrying where needed).

Answer: 3,651 x 7 = 25,557.

1(iii). Multiply mentally: 5,437 x 11.

Multiply 5,437 by 11.

Break 11 into 10 and 1.

5,437 x 1 = 5,437.

5,437 x 10 = 54,370.

Add the partial products: 5,437 + 54,370 = 59,807.

Answer: 5,437 x 11 = 59,807.

1(iv). Multiply mentally: 2,146 x 31.

Multiply 2,146 by 31.

Break 31 into 30 and 1.

2,146 x 1 = 2,146.

2,146 x 30 = 64,380.

Add the partial products: 2,146 + 64,380 = 66,526.

Answer: 2,146 x 31 = 66,526.

2(i). Multiply 4,620 x 8.

Multiply 4,620 by 8.

4,620 x 8 = 36,960 (multiply each digit by 8, carrying where needed).

Answer: 4,620 x 8 = 36,960.

2(ii). Multiply 3,657 x 5.

Multiply 3,657 by 5.

3,657 x 5 = 18,285 (multiply each digit by 5, carrying where needed).

Answer: 3,657 x 5 = 18,285.

2(iii). Multiply 9,884 x 6.

Multiply 9,884 by 6.

9,884 x 6 = 59,304 (multiply each digit by 6, carrying where needed).

Answer: 9,884 x 6 = 59,304.

2(iv). Multiply 7,378 x 9.

Multiply 7,378 by 9.

7,378 x 9 = 66,402 (multiply each digit by 9, carrying where needed).

Answer: 7,378 x 9 = 66,402.

3(i). Multiply 1,978 x 49.

Multiply 1,978 by 49.

Break 49 into 40 and 9.

1,978 x 9 = 17,802.

1,978 x 40 = 79,120.

Add the partial products: 17,802 + 79,120 = 96,922.

Answer: 1,978 x 49 = 96,922.

3(ii). Multiply 2,523 x 23.

Multiply 2,523 by 23.

Break 23 into 20 and 3.

2,523 x 3 = 7,569.

2,523 x 20 = 50,460.

Add the partial products: 7,569 + 50,460 = 58,029.

Answer: 2,523 x 23 = 58,029.

3(iii). Multiply 1,349 x 71.

Multiply 1,349 by 71.

Break 71 into 70 and 1.

1,349 x 1 = 1,349.

1,349 x 70 = 94,430.

Add the partial products: 1,349 + 94,430 = 95,779.

Answer: 1,349 x 71 = 95,779.

3(iv). Multiply 6,132 x 15.

Multiply 6,132 by 15.

Break 15 into 10 and 5.

6,132 x 5 = 30,660.

6,132 x 10 = 61,320.

Add the partial products: 30,660 + 61,320 = 91,980.

Answer: 6,132 x 15 = 91,980.

4(i). Divide 585 by 3.

Divide 585 by 3 by long division, starting from the highest place.

3 goes into 585 195 times (3 x 195 = 585).

Subtract: 585 - 585 = 0. Remainder = 0.

Answer: 585 / 3 = 195 exactly.

4(ii). Divide 816 by 4.

Divide 816 by 4 by long division, starting from the highest place.

4 goes into 816 204 times (4 x 204 = 816).

Subtract: 816 - 816 = 0. Remainder = 0.

Answer: 816 / 4 = 204 exactly.

4(iii). Divide 4,338 by 6.

Divide 4,338 by 6 by long division, starting from the highest place.

6 goes into 4,338 723 times (6 x 723 = 4,338).

Subtract: 4,338 - 4,338 = 0. Remainder = 0.

Answer: 4,338 / 6 = 723 exactly.

4(iv). Divide 1,998 by 9.

Divide 1,998 by 9 by long division, starting from the highest place.

9 goes into 1,998 222 times (9 x 222 = 1,998).

Subtract: 1,998 - 1,998 = 0. Remainder = 0.

Answer: 1,998 / 9 = 222 exactly.

5(i). Divide 345 by 19.

Divide 345 by 19 by long division, starting from the highest place.

19 goes into 345 18 times (19 x 18 = 342).

Subtract: 345 - 342 = 3. Remainder = 3.

Answer: 345 / 19 = 18, remainder 3.

5(ii). Divide 7,862 by 74.

Divide 7,862 by 74 by long division, starting from the highest place.

74 goes into 7,862 106 times (74 x 106 = 7,844).

Subtract: 7,862 - 7,844 = 18. Remainder = 18.

Answer: 7,862 / 74 = 106, remainder 18.

5(iii). Divide 5,600 by 20.

Divide 5,600 by 20 by long division, starting from the highest place.

20 goes into 5,600 280 times (20 x 280 = 5,600).

Subtract: 5,600 - 5,600 = 0. Remainder = 0.

Answer: 5,600 / 20 = 280 exactly.

5(iv). Divide 6,384 by 42.

Divide 6,384 by 42 by long division, starting from the highest place.

42 goes into 6,384 152 times (42 x 152 = 6,384).

Subtract: 6,384 - 6,384 = 0. Remainder = 0.

Answer: 6,384 / 42 = 152 exactly.

6. A factory produces 650 metres of cloth daily. How much cloth will be produced in 20 days?

Multiply 650 by 20.

Break 20 into 20 and 0.

650 x 0 = 0.

650 x 20 = 13,000.

Add the partial products: 0 + 13,000 = 13,000.

Answer: 650 x 20 = 13,000. The factory will produce 13,000 metres of cloth in 20 days.

7. In 45 relief camps, 2,244 blankets were distributed. How many blankets did each camp get, and how many were left over?

Divide 2,244 by 45 by long division, starting from the highest place.

45 goes into 2,244 49 times (45 x 49 = 2,205).

Subtract: 2,244 - 2,205 = 39. Remainder = 39.

Answer: 2,244 / 45 = 49, remainder 39. Each camp got 49 blankets and 39 blankets were left over.

8. Shoaib has 1,374 pages of scrap paper. She wants to make packets for her 32 students. How many pages will each packet have? How many extra pages will be left over?
Divide 1,374 by 32 by long division, starting from the highest place.

32 goes into 1,374 42 times (32 x 42 = 1,344).

Subtract: 1,374 - 1,344 = 30. Remainder = 30.

Answer: 1,374 / 32 = 42, remainder 30. Each packet will have 42 pages and 30 pages will be left over.

9. A team of 10 children won a science competition prize of Rs. 7,550, shared equally. How much will each child get?
Divide 7,550 by 10 by long division, starting from the highest place.

10 goes into 7,550 755 times (10 x 755 = 7,550).

Subtract: 7,550 - 7,550 = 0. Remainder = 0.

Answer: Rs. 7,550 / 10 = Rs. 755. Each child will get Rs. 755.

Unit 4: Multiples and Factors

Unit 4: Multiples and Factors. Divisibility rules for 2, 3, 5 and 10, prime and composite numbers, multiples, common multiples, factors, factor pairs and common factors.

Exercise 4.1

1. Divisible by 2? 26, 95, 82, 219, 264, 1,340, 2,351, 1,278, 4,417, 9,750.

(i) 26: divisible, because the ones digit is even (0, 2, 4, 6 or 8) (ii) 95: not divisible, because the ones digit is odd (iii) 82: divisible, because the ones digit is even (0, 2, 4, 6 or 8) (iv) 219: not divisible, because the ones digit is odd (v) 264: divisible, because the ones digit is even (0, 2, 4, 6 or 8) (vi) 1,340: divisible, because the ones digit is even (0, 2, 4, 6 or 8) (vii) 2,351: not divisible, because the ones digit is odd (viii) 1,278: divisible, because the ones digit is even (0, 2, 4, 6 or 8) (ix) 4,417: not divisible, because the ones digit is odd (x) 9,750: divisible, because the ones digit is even (0, 2, 4, 6 or 8).

2. Divisible by 3? 123, 472, 891, 9,381, 7,234, 7,437, 1,359, 2,348.

(i) 123: divisible, because the sum of its digits is 6 (a multiple of 3) (ii) 472: not divisible, because the sum of its digits is 13 (not a multiple of 3) (iii) 891: divisible, because the sum of its digits is 18 (a multiple of 3) (iv) 9,381: divisible, because the sum of its digits is 21 (a multiple of 3) (v) 7,234: not divisible, because the sum of its digits is 16 (not a multiple of 3) (vi) 7,437: divisible, because the

sum of its digits is 21 (a multiple of 3) (vii) 1,359: divisible, because the sum of its digits is 18 (a multiple of 3) (viii) 2,348: not divisible, because the sum of its digits is 17 (not a multiple of 3).

3. Divisible by 5? 211, 415, 970, 1,235, 9,760, 7,777, 1,375, 3,551.
(i) 211: not divisible, because the ones digit is neither 0 nor 5 (ii) 415: divisible, because the ones digit is 0 or 5 (iii) 970: divisible, because the ones digit is 0 or 5 (iv) 1,235: divisible, because the ones digit is 0 or 5 (v) 9,760: divisible, because the ones digit is 0 or 5 (vi) 7,777: not divisible, because the ones digit is neither 0 nor 5 (vii) 1,375: divisible, because the ones digit is 0 or 5 (viii) 3,551: not divisible, because the ones digit is neither 0 nor 5.

4. Divisible by 10? 510, 703, 919, 5,670, 4,682, 5,720, 1,001, 3,910.
(i) 510: divisible, because the ones digit is 0 (ii) 703: not divisible, because the ones digit is not 0 (iii) 919: not divisible, because the ones digit is not 0 (iv) 5,670: divisible, because the ones digit is 0 (v) 4,682: not divisible, because the ones digit is not 0 (vi) 5,720: divisible, because the ones digit is 0 (vii) 1,001: not divisible, because the ones digit is not 0 (viii) 3,910: divisible, because the ones digit is 0.

5. Complete the number flower: sort these numbers into the flowers for 2, 3 and 5: 12, 15, 18, 25, 30, 32, 34, 36, 45, 165, 84, 86, 90, 95, 96, 100, 105, 120, 135.
Divisible by 2: 12, 18, 30, 32, 34, 36, 84, 86, 90, 96, 100, 120.
Divisible by 3: 12, 15, 18, 30, 36, 45, 165, 84, 90, 96, 105, 120,

135. Divisible by 5: 15, 25, 30, 45, 165, 90, 95, 100, 105, 120, 135. A number can appear in more than one flower, for example 30 is divisible by 2, 3 and 5.

6. Write the numbers greater than 999 that are divisible by 2 and 5 in the balloons. The printed balloons show 1,000, 1,010, 1,020, 1,050, 1,070, 1,090, 1,110 and 1,130 with gaps between them.
A number divisible by both 2 and 5 ends in 0. Counting by tens from 1,000, the missing balloons are 1,030, 1,040, 1,060, 1,080, 1,100, 1,120, 1,140 and 1,150. In order the full run is 1,000, 1,010, 1,020, 1,030, 1,040, 1,050, 1,060, 1,070, 1,080, 1,090, 1,100, 1,110, 1,120, 1,130, 1,140, 1,150.

Exercise 4.2

1. Colour the prime numbers yellow and the composite numbers blue (numbers 1 to 50). Which are prime and which are composite?
Prime numbers up to 50: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47. Composite numbers from 4 to 50: all the other numbers except 1, for example 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 30, 32, 33, 34, 35, 36, 38, 39, 40, 42, 44, 45, 46, 48, 49, 50. The number 1 is neither prime nor composite, because it has only one factor.

2. How many composite numbers and how many prime numbers are there up to 50?
There are 15 prime numbers up to 50 and 34 composite numbers up to 50. Together with 1, which is neither, that makes 50 numbers.

Exercise 4.3

1. List the first ten multiples of 2.

Multiples of 2: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. Multiply {n} by 1, 2, 3 and so on.

1. List the first ten multiples of 3.

Multiples of 3: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. Multiply {n} by 1, 2, 3 and so on.

1. List the first ten multiples of 4.

Multiples of 4: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40. Multiply {n} by 1, 2, 3 and so on.

1. List the first ten multiples of 5.

Multiples of 5: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50. Multiply {n} by 1, 2, 3 and so on.

1. List the first ten multiples of 6.

Multiples of 6: 6, 12, 18, 24, 30, 36, 42, 48, 54, 60. Multiply {n} by 1, 2, 3 and so on.

1. List the first ten multiples of 7.

Multiples of 7: 7, 14, 21, 28, 35, 42, 49, 56, 63, 70. Multiply {n} by 1, 2, 3 and so on.

1. List the first ten multiples of 8.

Multiples of 8: 8, 16, 24, 32, 40, 48, 56, 64, 72, 80. Multiply {n} by 1, 2, 3 and so on.

1. List the first ten multiples of 9.

Multiples of 9: 9, 18, 27, 36, 45, 54, 63, 72, 81, 90. Multiply {n} by 1, 2, 3 and so on.

1. List the first ten multiples of 10.

Multiples of 10: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. Multiply {n} by 1, 2, 3 and so on.

2(a). Observe the pattern and write the next five multiples: 6, 12, 18, 24.

The pattern adds 6 each time. The next five multiples are 30, 36, 42, 48, 54.

2(b). Observe the pattern and write the next five multiples: 9, 18, 27, 36.

The pattern adds 9 each time. The next five multiples are 45, 54, 63, 72, 81.

2(c). Observe the pattern and write the next five multiples: 12, 24, 36, 48.

The pattern adds 12 each time. The next five multiples are 60, 72, 84, 96, 108.

2(d). Observe the pattern and write the next five multiples: 15, 30, 45, 60.

The pattern adds 15 each time. The next five multiples are 75, 90, 105, 120, 135.

2(e). Observe the pattern and write the next five multiples: 17, 34, 51, 68.

The pattern adds 17 each time. The next five multiples are 85, 102, 119, 136, 153.

Exercise 4.4

1(a). Write the first ten multiples of 6 and 9 and find the common multiples.

Multiples of 6: 6, 12, 18, 24, 30, 36, 42, 48, 54, 60. Multiples of 9: 9, 18, 27, 36, 45, 54, 63, 72, 81, 90. Common multiples: 18, 36, 54.

1(b). Write the first ten multiples of 4 and 8 and find the common multiples.

Multiples of 4: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40. Multiples of 8: 8, 16, 24, 32, 40, 48, 56, 64, 72, 80. Common multiples: 8, 16, 24.

1(c). Write the first ten multiples of 10 and 15 and find the common multiples.

Multiples of 10: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. Multiples of 15: 15, 30, 45, 60, 75, 90, 105, 120, 135, 150. Common multiples: 30, 60, 90.

1(d). Write the first ten multiples of 3 and 4 and find the common multiples.

Multiples of 3: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. Multiples of 4: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40. Common multiples: 12, 24.

2(a). Write the common multiples of 3, 5, 15.

The common multiples are 15, 30, 45. These are the multiples of the lowest common multiple, 15.

2(b). Write the common multiples of 3, 6, 9.

The common multiples are 18, 36, 54. These are the multiples of the lowest common multiple, 18.

2(c). Write the common multiples of 4, 6, 12.

The common multiples are 12, 24, 36. These are the multiples of the lowest common multiple, 12.

Exercise 4.5

1(a). Fill in the missing factors of 6 (given: 1, 2).

All factors of 6 are 1, 2, 3, 6. The missing factors are 3, 6.

1(b). Fill in the missing factors of 10 (given: 1, 2).

All factors of 10 are 1, 2, 5, 10. The missing factors are 5, 10.

1(c). Fill in the missing factors of 15 (given: 1, 3).

All factors of 15 are 1, 3, 5, 15. The missing factors are 5, 15.

1(d). Fill in the missing factors of 18 (given: 1, 2, 3).

All factors of 18 are 1, 2, 3, 6, 9, 18. The missing factors are 6, 9, 18.

1(e). Fill in the missing factors of 32 (given: 1, 2).

All factors of 32 are 1, 2, 4, 8, 16, 32. The missing factors are 4, 8,

16, 32.

1(f). Fill in the missing factors of 50 (given: 1, 2, 4).

All factors of 50 are 1, 2, 5, 10, 25, 50. The missing factors are 5, 10, 25, 50.

2(a). List all the pair factors of 24.

Pair factors of 24: (1, 24), (2, 12), (3, 8), (4, 6). Each pair multiplies to 24.

2(b). List all the pair factors of 28.

Pair factors of 28: (1, 28), (2, 14), (4, 7). Each pair multiplies to 28.

2(c). List all the pair factors of 36.

Pair factors of 36: (1, 36), (2, 18), (3, 12), (4, 9), (6, 6). Each pair multiplies to 36.

2(d). List all the pair factors of 64.

Pair factors of 64: (1, 64), (2, 32), (4, 16), (8, 8). Each pair multiplies to 64.

2(e). List all the pair factors of 48.

Pair factors of 48: (1, 48), (2, 24), (3, 16), (4, 12), (6, 8). Each pair multiplies to 48.

2(f). List all the pair factors of 78.

Pair factors of 78: (1, 78), (2, 39), (3, 26), (6, 13). Each pair multiplies to 78.

2(g). List all the pair factors of 96.

Pair factors of 96: (1, 96), (2, 48), (3, 32), (4, 24), (6, 16), (8, 12). Each pair multiplies to 96.

3(i). Find the common factors of 18 and 24.

Factors of 18: 1, 2, 3, 6, 9, 18. Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Common factors: 1, 2, 3, 6.

3(ii). Find the common factors of 20 and 30.

Factors of 20: 1, 2, 4, 5, 10, 20. Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. Common factors: 1, 2, 5, 10.

3(iii). Find the common factors of 56 and 63.

Factors of 56: 1, 2, 4, 7, 8, 14, 28, 56. Factors of 63: 1, 3, 7, 9, 21, 63. Common factors: 1, 7.

3(iv). Find the common factors of 45 and 75.

Factors of 45: 1, 3, 5, 9, 15, 45. Factors of 75: 1, 3, 5, 15, 25, 75. Common factors: 1, 3, 5, 15.

3(v). Find the common factors of 60 and 90.

Factors of 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. Factors of 90: 1, 2, 3, 5, 6, 9, 10, 15, 18, 30, 45, 90. Common factors: 1, 2, 3, 5, 6, 10, 15, 30.

Exercise 4.6

1(a). Write all the factors of 16.

Factors of 16: 1, 2, 4, 8, 16. These are all the numbers that divide 16 exactly.

1(b). Write all the factors of 28.

Factors of 28: 1, 2, 4, 7, 14, 28. These are all the numbers that divide 28 exactly.

1(c). Write all the factors of 36.

Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. These are all the numbers that divide 36 exactly.

1(d). Write all the factors of 48.

Factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. These are all the numbers that divide 48 exactly.

1(e). Write all the factors of 72.

Factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72. These are all the numbers that divide 72 exactly.

2(a). Write the first 8 multiples of 7.

Multiples of 7: 7, 14, 21, 28, 35, 42, 49, 56.

2(b). Write the first 8 multiples of 8.

Multiples of 8: 8, 16, 24, 32, 40, 48, 56, 64.

2(c). Write the first 8 multiples of 9.

Multiples of 9: 9, 18, 27, 36, 45, 54, 63, 72.

2(d). Write the first 8 multiples of 12.

Multiples of 12: 12, 24, 36, 48, 60, 72, 84, 96.

Review Exercise 4

1. Use the divisibility rules to complete the table for 212, 986, 6409, 6600, 81810, 59012, 53800, 21034 (divisible by 2, 3, 5, 10).

212: by 2 yes, by 3 no, by 5 no, by 10 no 986: by 2 yes, by 3 no, by 5 no, by 10 no 6,409: by 2 no, by 3 no, by 5 no, by 10 no 6,600: by 2 yes, by 3 yes, by 5 yes, by 10 yes 81,810: by 2 yes, by 3 yes, by 5 yes, by 10 yes 59,012: by 2 yes, by 3 no, by 5 no, by 10 no 53,800: by 2 yes, by 3 no, by 5 yes, by 10 yes 21,034: by 2 yes, by 3 no, by 5 no, by 10 no. Note: the book's printed key wrongly marks 986 as divisible by 3; its digits add to 23, which is not a multiple of 3, so 986 is not divisible by 3.

2(i). Write the first ten multiples of 5, 9 and find the common multiples.

Common multiples: 45, 90, 135. The lowest common multiple is 45.

2(ii). Write the first ten multiples of 12, 15 and find the common multiples.

Common multiples: 60, 120, 180. The lowest common multiple is 60.

2(iii). Write the first ten multiples of 20, 30 and find the common multiples.

Common multiples: 60, 120, 180. The lowest common multiple is 60.

2(iv). Write the first ten multiples of 6, 9, 12 and find the common multiples.

Common multiples: 36, 72, 108. The lowest common multiple is 36.

2(v). Write the first ten multiples of 5, 10, 15 and find the common multiples.

Common multiples: 30, 60, 90. The lowest common multiple is 30.

3(i). Find the common factors of 8 and 12.

Factors of 8: 1, 2, 4, 8. Factors of 12: 1, 2, 3, 4, 6, 12. Common factors: 1, 2, 4.

3(ii). Find the common factors of 28 and 42.

Factors of 28: 1, 2, 4, 7, 14, 28. Factors of 42: 1, 2, 3, 6, 7, 14, 21, 42. Common factors: 1, 2, 7, 14.

3(iii). Find the common factors of 48 and 72.

Factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. Factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72. Common factors: 1, 2, 3, 4, 6, 8, 12, 24.

Unit 5: Fractions

Unit 5: Fractions. Proper, improper and mixed numbers, conversions, comparing and ordering, adding and subtracting like and unlike fractions, multiplying and dividing fractions, and word problems. Fractions are written as p/q and mixed numbers as w p/q.

Exercise 5.1

1. Sort out the proper fractions, improper fractions and mixed numbers separately.

Proper fractions (numerator smaller than denominator): 9/10, 8/11, 3/7, 11/72, 19/21, 11/12, 35/92. Improper fractions (numerator greater than denominator): 15/8, 17/9, 25/18, 55/26, 62/32, 19/17, 33/26, 39/16. Mixed numbers (a whole number with a fraction): 3 27/37, 8 7/37, 5 23/27, 5 2/5, 9 12/25, 9 1/5, 6 24/41, 3 12/35.

Exercise 5.2

1(a). Convert 8/5 into a mixed number.

Divide 8 by 5: 5 goes 1 times with remainder 3. So 8/5 = 1 3/5.

1(b). Convert 11/5 into a mixed number.

Divide 11 by 5: 5 goes 2 times with remainder 1. So 11/5 = 2 1/5.

1(c). Convert 13/10 into a mixed number.

Divide 13 by 10: 10 goes 1 times with remainder 3. So 13/10 = 1 3/10.

1(d). Convert 20/9 into a mixed number.

Divide 20 by 9: 9 goes 2 times with remainder 2. So 20/9 = 2 2/9.

1(e). Convert 15/2 into a mixed number.

Divide 15 by 2: 2 goes 7 times with remainder 1. So 15/2 = 7 1/2.

2(a). Convert 2 3/5 into an improper fraction.

Multiply the whole number by the denominator and add the numerator: 2 x 5 + 3 = 13. So 2 3/5 = 13/5.

2(b). Convert 3 8/11 into an improper fraction.

Multiply the whole number by the denominator and add the numerator: 3 x 11 + 8 = 41. So 3 8/11 = 41/11.

2(c). Convert 5 6/7 into an improper fraction.

Multiply the whole number by the denominator and add the numerator: 5 x 7 + 6 = 41. So 5 6/7 = 41/7.

2(d). Convert 3 7/13 into an improper fraction.

Multiply the whole number by the denominator and add the numerator: 3 x 13 + 7 = 46. So 3 7/13 = 46/13.

2(e). Convert 2 11/19 into an improper fraction.

Multiply the whole number by the denominator and add the numerator: 2 x 19 + 11 = 49. So 2 11/19 = 49/19.

Exercise 5.3

1(a). Compare 1/2 and 3/6 using <, > or =.

1/2 = 3/6. In decimals this is about 0.500 and 0.500, so the sign is =.

1(b). Compare 2/9 and 5/9 using <, > or =.

2/9 < 5/9. In decimals this is about 0.222 and 0.556, so the sign is <.

1(c). Compare 4/5 and 9/10 using <, > or =.

4/5 < 9/10. In decimals this is about 0.800 and 0.900, so the sign is <.

1(d). Compare 6/12 and 1/4 using <, > or =.

6/12 > 1/4. In decimals this is about 0.500 and 0.250, so the sign is >.

1(e). Compare 1/3 and 1/4 using <, > or =.

1/3 > 1/4. In decimals this is about 0.333 and 0.250, so the sign is >.

1(f). Compare 4/11 and 5/7 using <, > or =.

4/11 < 5/7. In decimals this is about 0.364 and 0.714, so the sign is <.

2(i). Write in ascending and descending order: 3/5, 3/9, 3/7.

Ascending order: 3/9, 3/7, 3/5. Descending order: 3/5, 3/7, 3/9. Compare by converting to a common denominator or to decimals.

2(ii). Write in ascending and descending order: 3/4, 1/3, 6/7.

Ascending order: 1/3, 3/4, 6/7. Descending order: 6/7, 3/4, 1/3. Compare by converting to a common denominator or to decimals.

2(iii). Write in ascending and descending order: 3/5, 2/10, 4/15.
Ascending order: 2/10, 4/15, 3/5. Descending order: 3/5, 4/15, 2/10. Compare by converting to a common denominator or to decimals.

Exercise 5.4

1(i). Add 2/9 + 5/9 and write the answer in mixed number (where it applies).
2/9 + 5/9 = 7/9. Convert to a common denominator first, add the numerators, and simplify.

1(ii). Add 4/5 + 8/15 and write the answer in mixed number (where it applies).
4/5 + 8/15 = 1 1/3. Convert to a common denominator first, add the numerators, and simplify.

1(iii). Add 4/13 + 7/13 and write the answer in mixed number (where it applies).
4/13 + 7/13 = 11/13. Convert to a common denominator first, add the numerators, and simplify.

1(iv). Add 22/9 + 1/27 and write the answer in mixed number (where it applies).
22/9 + 1/27 = 2 13/27. Convert to a common denominator first, add the numerators, and simplify.

1(v). Add 11/25 + 8/25 and write the answer in mixed number (where it applies).
11/25 + 8/25 = 19/25. Convert to a common denominator first, add the numerators, and simplify.

1(vi). Add 1/21 + 5/3 and write the answer in mixed number (where it applies).
1/21 + 5/3 = 1 5/7. Convert to a common denominator first, add the numerators, and simplify.

1(vii). Add 3/4 + 7/12 and write the answer in mixed number (where it applies).
3/4 + 7/12 = 1 1/3. Convert to a common denominator first, add the numerators, and simplify.

1(viii). Add 4/22 + 8/22 and write the answer in mixed number (where it applies).
4/22 + 8/22 = 6/11. Convert to a common denominator first, add the numerators, and simplify.

2(i). Subtract 5/12 from 7/12 and write the answer in mixed number (where it applies).
7/12 - 5/12 = 1/6. Convert to a common denominator first, subtract the numerators, and simplify.

2(ii). Subtract 11/21 from 5/7 and write the answer in mixed number (where it applies).
5/7 - 11/21 = 4/21. Convert to a common denominator first,

subtract the numerators, and simplify.

2(iii). Subtract 19/36 from 11/18 and write the answer in mixed number (where it applies).
11/18 - 19/36 = 1/12. Convert to a common denominator first, subtract the numerators, and simplify.

2(iv). Subtract 8/25 from 9/5 and write the answer in mixed number (where it applies).
9/5 - 8/25 = 1 12/25. Convert to a common denominator first, subtract the numerators, and simplify.

2(v). Subtract 13/10 from 17/10 and write the answer in mixed number (where it applies).
17/10 - 13/10 = 2/5. Convert to a common denominator first, subtract the numerators, and simplify.

2(vi). Subtract 5/22 from 4/11 and write the answer in mixed number (where it applies).
4/11 - 5/22 = 3/22. Convert to a common denominator first, subtract the numerators, and simplify.

2(vii). Subtract 2/9 from 7/9 and write the answer in mixed number (where it applies).
7/9 - 2/9 = 5/9. Convert to a common denominator first, subtract the numerators, and simplify.

2(viii). Subtract 1/3 from 19/33 and write the answer in mixed

number (where it applies).

19/33 - 1/3 = 8/33. Convert to a common denominator first, subtract the numerators, and simplify.

Exercise 5.5

1(i). Multiply: 6/7 x 5/6.

Convert any mixed numbers to improper fractions, multiply numerators together and denominators together, then simplify. 6/7 x 5/6 = 5/7.

1(ii). Multiply: 9/11 x 5/7.

Convert any mixed numbers to improper fractions, multiply numerators together and denominators together, then simplify. 9/11 x 5/7 = 45/77.

1(iii). Multiply: 3/17 x 3 3/4.

Convert any mixed numbers to improper fractions, multiply numerators together and denominators together, then simplify. 3/17 x 3 3/4 = 45/68.

1(iv). Multiply: 7 1/7 x 1 5/8.

Convert any mixed numbers to improper fractions, multiply numerators together and denominators together, then simplify. 7 1/7 x 1 5/8 = 11 17/28.

1(v). Multiply: 4/3 x 1/4 x 7 7/10.

Convert any mixed numbers to improper fractions, multiply numerators together and denominators together, then simplify. 4/3

x 1/4 x 7 7/10 = 2 17/30.

1(vi). Multiply: 9/12 x 3 8/11.

Convert any mixed numbers to improper fractions, multiply numerators together and denominators together, then simplify. 9/12 x 3 8/11 = 2 35/44.

1(vii). Multiply: 2/9 x 1 5/6.

Convert any mixed numbers to improper fractions, multiply numerators together and denominators together, then simplify. 2/9 x 1 5/6 = 11/27.

1(viii). Multiply: 11/12 x 7.

Convert any mixed numbers to improper fractions, multiply numerators together and denominators together, then simplify. 11/12 x 7 = 6 5/12.

1(ix). Multiply: 8/5 x 23.

Convert any mixed numbers to improper fractions, multiply numerators together and denominators together, then simplify. 8/5 x 23 = 36 4/5.

1(x). Multiply: (first factor missing in the printed book) x 7 5/41 x 11.

As printed, item (x) is missing its first factor, so it cannot be solved as written; nothing has been guessed here. The printed part alone is 7 5/41 x 11 = 292/41 x 11 = 3,212/41 = 78 14/41.

2(i). Divide: 7/8 / 35/48.

Change division to multiplication by the reciprocal of the second fraction, then multiply and simplify. 7/8 / 35/48 = 1 1/5.

2(ii). Divide: 9/16 / 22/32.

Change division to multiplication by the reciprocal of the second fraction, then multiply and simplify. 9/16 / 22/32 = 9/11.

2(iii). Divide: 14/13 / 56/39.

Change division to multiplication by the reciprocal of the second fraction, then multiply and simplify. 14/13 / 56/39 = 3/4.

2(iv). Divide: 10/75 / 20/225.

Change division to multiplication by the reciprocal of the second fraction, then multiply and simplify. 10/75 / 20/225 = 1 1/2.

2(v). Divide: 3/2 / 31.

Change division to multiplication by the reciprocal of the second fraction, then multiply and simplify. 3/2 / 31 = 3/62.

2(vi). Divide: 29/33 / 2.

Change division to multiplication by the reciprocal of the second fraction, then multiply and simplify. 29/33 / 2 = 29/66.

2(vii). Divide: 5 11/17 / 16.

Change division to multiplication by the reciprocal of the second fraction, then multiply and simplify. 5 11/17 / 16 = 6/17.

Exercise 5.6

1. A plant grows 1/6 inch in one week and 1/12 inch in another week. How much did it grow in total?

Total growth = 1/6 + 1/12 = 2/12 + 1/12 = 3/12 = 1/4 inch. The plant grew 1/4 inch in total.

2. A tailor has 3/4 metre of cloth. After using 1/8 metre, how much cloth is left?

Cloth left = 3/4 - 1/8 = 6/8 - 1/8 = 5/8 metre. The tailor has 5/8 metre of cloth left.

3. Shazia has 2 1/2 m long ribbon. She cut off 7/9 of ribbon for packing a gift. How much ribbon is left?

Ribbon left = 2 1/2 - 7/9 = 5/2 - 7/9 = 45/18 - 14/18 = 31/18 = 1 13/18 m. Shazia has 1 13/18 m of ribbon left.

4. Farooq has 2 pizzas and wants to share them equally among 4 friends. How much pizza does each friend get?

Each friend gets 2 / 4 = 1/2 pizza. Note: the question is printed with a gap after the number 2; it is solved here as 2 pizzas, exactly as the printed number reads.

Review Exercise 5

1. Sort the proper and improper fractions: 9/10, 8/11, 3/7, 15/8, 17/3, 25/18, 27/35, 11/72, 19/21, 55/26, 62/31.

Proper fractions: 9/10, 8/11, 3/7, 27/35, 11/72, 19/21. Improper fractions: 15/8, 17/3, 25/18, 55/26, 62/31.

2. Sort out the mixed numbers: 5 1/7, 12/25, 33/26, 2 1/5, 24/40, 3 6/13, 39/16, 12/11, 92/35, 3 55/78.
Mixed numbers: 5 1/7, 2 1/5, 3 6/13, 3 55/78. The other items are ordinary fractions, not mixed numbers.

3(i). Convert 28/8 to a mixed fraction in the lowest form.
Divide 28 by 8: 8 goes 3 times with remainder 4. So 28/8 = 3 1/2.

3(ii). Convert 26/9 to a mixed fraction in the lowest form.
Divide 26 by 9: 9 goes 2 times with remainder 8. So 26/9 = 2 8/9.

3(iii). Convert 39/9 to a mixed fraction in the lowest form.
Divide 39 by 9: 9 goes 4 times with remainder 3. So 39/9 = 4 1/3.

3(iv). Convert 38/3 to a mixed fraction in the lowest form.
Divide 38 by 3: 3 goes 12 times with remainder 2. So 38/3 = 12 2/3.

3(v). Convert 79/5 to a mixed fraction in the lowest form.
Divide 79 by 5: 5 goes 15 times with remainder 4. So 79/5 = 15 4/5.

3(vi). Convert 87/9 to a mixed fraction in the lowest form.
Divide 87 by 9: 9 goes 9 times with remainder 6. So 87/9 = 9 2/3.

4(i). Convert 8 1/2 into an improper fraction.
8 x 2 + 1 = 17. So 8 1/2 = 17/2.

4(ii). Convert 4 3/10 into an improper fraction.

4 x 10 + 3 = 43. So 4 3/10 = 43/10.

4(iii). Convert 8 4/5 into an improper fraction.

8 x 5 + 4 = 44. So 8 4/5 = 44/5.

4(iv). Convert 7 9/10 into an improper fraction.

7 x 10 + 9 = 79. So 7 9/10 = 79/10.

4(v). Convert 3 5/8 into an improper fraction.

3 x 8 + 5 = 29. So 3 5/8 = 29/8.

4(vi). Convert 9 1/8 into an improper fraction.

9 x 8 + 1 = 73. So 9 1/8 = 73/8.

5(i). Add 6/13 + 8/13 and write it in the lowest form.

6/13 + 8/13 = 1 1/13. Add the numerators over the common denominator and simplify.

5(ii). Add 12/17 + 14/17 and write it in the lowest form.

12/17 + 14/17 = 1 9/17. Add the numerators over the common denominator and simplify.

5(iii). Add 8/9 + 11/9 and write it in the lowest form.

8/9 + 11/9 = 2 1/9. Add the numerators over the common denominator and simplify.

5(iv). Add 15/16 + 3/8 and write it in the lowest form.

15/16 + 3/8 = 1 5/16. Add the numerators over the common denominator and simplify.

6(i). Subtract 3/10 from 5/10 and write it in the lowest form.

5/10 - 3/10 = 1/5. Convert to a common denominator, subtract, and simplify.

6(ii). Subtract 4/12 from 7/12 and write it in the lowest form.

7/12 - 4/12 = 1/4. Convert to a common denominator, subtract, and simplify.

6(iii). Subtract 3/16 from 11/8 and write it in the lowest form.

11/8 - 3/16 = 1 3/16. Convert to a common denominator, subtract, and simplify.

6(iv). Subtract 8/25 from 9/5 and write it in the lowest form.

9/5 - 8/25 = 1 12/25. Convert to a common denominator, subtract, and simplify.

7. A rope was cut, 2/7 metre, then another 1/7 metre. How much rope was cut in total?

Total cut = 2/7 + 1/7 = 3/7 metre. Altogether 3/7 metre of rope was cut.

8. If the weight of 5 packets of sugar is 7 1/2 kg, what will be the weight of 1 packet of sugar?

One packet = 7 1/2 / 5 = 15/2 / 5 = 15/10 = 3/2 = 1 1/2 kg. One packet weighs 1 1/2 kg.

9. A man travelled 95 1/10 km in one day. He travelled 61 1/5 km by car and the rest by scooter. How far did he travel by scooter?
Distance by scooter = 95 1/10 - 61 1/5 = 951/10 - 612/10 = 339/10 = 33 9/10 km. He travelled 33 9/10 km by scooter.

10. A tin contains 18 2/3 litres of oil. 25/6 litres leaked out. How much oil is left in the tin?
Oil left = 18 2/3 - 25/6 = 56/3 - 25/6 = 112/6 - 25/6 = 87/6 = 29/2 = 14 1/2 litres. There are 14 1/2 litres of oil left.

Unit 6: Decimals

Unit 6: Decimals. Place value of decimals, percent, converting fractions and decimals, comparing and ordering decimals, rounding decimals, adding, subtracting, multiplying and dividing decimals, order of operations (DMAS), and word problems. Note: two worked examples in the book itself swap values (427/100 and 4321/1000); the correct values are used here: 427/100 = 4.27 and 4321/1000 = 4.321.

Exercise 6.1

1. Identify the values and write the places of tenths and hundredths: (i) 3/10 = 0.3; (ii) 45/100 = 0.45; (iii) 19/100 = 0.19; (iv) 7/10 = 0.7; (v) 5/100 = 0.05.

(i) 3/10 = 0.3: Tenths 3, Hundredths none. (ii) 45/100 = 0.45: Tenths 4, Hundredths 5. (iii) 19/100 = 0.19: Tenths 1, Hundredths 9. (iv) 7/10 = 0.7: Tenths 7, Hundredths none. (v) 5/100 = 0.05: Tenths 0, Hundredths 5.

2(i). How many tenths, hundredths and thousandths are in 0.578?

0.578 has 5 tenths, 7 hundredths and 8 thousandths. Read the digits after the decimal point in order.

2(ii). How many tenths, hundredths and thousandths are in 0.345?

0.345 has 3 tenths, 4 hundredths and 5 thousandths. Read the digits after the decimal point in order.

2(iii). How many tenths, hundredths and thousandths are in 0.934?

0.934 has 9 tenths, 3 hundredths and 4 thousandths. Read the digits after the decimal point in order.

2(iv). How many tenths, hundredths and thousandths are in 0.117?

0.117 has 1 tenths, 1 hundredths and 7 thousandths. Read the digits after the decimal point in order.

2(v). How many tenths, hundredths and thousandths are in 0.421?

0.421 has 4 tenths, 2 hundredths and 1 thousandths. Read the digits after the decimal point in order.

2(vi). How many tenths, hundredths and thousandths are in 0.264?

0.264 has 2 tenths, 6 hundredths and 4 thousandths. Read the digits after the decimal point in order.

3(i). Write the percent for 20 parts out of 100.

20 parts out of 100 = 20%, because percent means parts per hundred.

3(ii). Write the percent for 15 parts out of 100.

15 parts out of 100 = 15%, because percent means parts per hundred.

3(iii). Write the percent for 45 parts out of 100.

45 parts out of 100 = 45%, because percent means parts per hundred.

3(iv). Write the percent for 80 parts out of 100.

80 parts out of 100 = 80%, because percent means parts per hundred.

3(v). Write the percent for 98 parts out of 100.

98 parts out of 100 = 98%, because percent means parts per hundred.

Exercise 6.2

1(i). Convert 1/10 into a decimal.

Change 1/10 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 1/10 = 0.1.

1(ii). Convert 9/5 into a decimal.

Change 9/5 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 9/5 = 1.8.

1(iii). Convert 97/100 into a decimal.

Change 97/100 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 97/100 = 0.97.

1(iv). Convert 72/50 into a decimal.

Change 72/50 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 72/50 = 1.44.

1(v). Convert 333/250 into a decimal.

Change 333/250 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 333/250 = 1.332.

1(vi). Convert 32/500 into a decimal.

Change 32/500 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 32/500 = 0.064.

1(vii). Convert 2354/1000 into a decimal.

Change 2354/1000 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 2354/1000 = 2.354.

1(viii). Convert 23/75 into a decimal.

23/75 cannot be changed into an exact fraction with denominator 10, 100 or 1000, because 75 does not divide any of them exactly. By division, 23 / 75 = 0.3066..., about 0.307 to three decimal places. The book's key gives 0.306, which is the value cut off after three decimal places.

1(ix). Convert 7/5 into a decimal.

Change 7/5 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 7/5 = 1.4.

2(i). Express 0.7 as a common fraction.

Write the digits after the decimal point over the matching place value and simplify. 0.7 = 7/10.

2(ii). Express 8.7 as a common fraction.

Write the digits after the decimal point over the matching place value and simplify. 8.7 = 87/10.

2(iii). Express 1.29 as a common fraction.

Write the digits after the decimal point over the matching place value and simplify. 1.29 = 129/100.

2(iv). Express 3.74 as a common fraction.

Write the digits after the decimal point over the matching place value and simplify. 3.74 = 187/50.

2(v). Express 7.297 as a common fraction.

Write the digits after the decimal point over the matching place value and simplify. 7.297 = 7297/1000.

2(vi). Express 0.143 as a common fraction.

Write the digits after the decimal point over the matching place value and simplify. 0.143 = 143/1000.

3. Convert a quarter (1/4), a half (1/2) and three quarters (3/4) to decimal equivalents.

1/4 = 25/100 = 0.25. 1/2 = 50/100 = 0.5. 3/4 = 75/100 = 0.75.

Exercise 6.3

1(i). Compare 3.48 and 4.23.

3.48 < 4.23. Compare the ones first, then the tenths, then the hundredths, until the digits differ.

1(ii). Compare 1.23 and 1.13.

1.23 > 1.13. Compare the ones first, then the tenths, then the hundredths, until the digits differ.

1(iii). Compare 4.67 and 4.68.

4.67 < 4.68. Compare the ones first, then the tenths, then the hundredths, until the digits differ.

1(iv). Compare 3.36 and 3.36.

3.36 = 3.36. Compare the ones first, then the tenths, then the hundredths, until the digits differ.

1(v). Compare 6.85 and 6.95.

6.85 < 6.95. Compare the ones first, then the tenths, then the hundredths, until the digits differ.

1(vi). Compare 9.27 and 9.25.

9.27 > 9.25. Compare the ones first, then the tenths, then the hundredths, until the digits differ.

2(i). Order in ascending and descending order: 3.25, 3.26, 3.25, 3.21, 3.24.

Ascending order: 3.21, 3.24, 3.25, 3.25, 3.26. Descending order: 3.26, 3.25, 3.25, 3.24, 3.21.

2(ii). Order in ascending and descending order: 4.25, 3.62, 4.91, 3.25, 8.42.

Ascending order: 3.25, 3.62, 4.25, 4.91, 8.42. Descending order: 8.42, 4.91, 4.25, 3.62, 3.25.

2(iii). Order in ascending and descending order: 9.21, 9.23, 9.61,

9.54, 9.35.

Ascending order: 9.21, 9.23, 9.35, 9.54, 9.61. Descending order: 9.61, 9.54, 9.35, 9.23, 9.21.

2(iv). Order in ascending and descending order: 8.15, 8.16, 8.11, 8.1, 8.13.

Ascending order: 8.1, 8.11, 8.13, 8.15, 8.16. Descending order: 8.16, 8.15, 8.13, 8.11, 8.1.

3(i). Round 4.567 to the nearest whole number, to tenths and to hundredths.

Whole number: 5. Tenths: 4.6. Hundredths: 4.57. Look at the next digit to the right: 5 or more rounds up, less than 5 stays.

3(ii). Round 5.321 to the nearest whole number, to tenths and to hundredths.

Whole number: 5. Tenths: 5.3. Hundredths: 5.32. Look at the next digit to the right: 5 or more rounds up, less than 5 stays.

3(iii). Round 11.43 to the nearest whole number, to tenths and to hundredths.

Whole number: 11. Tenths: 11.4. Hundredths: 11.43. Look at the next digit to the right: 5 or more rounds up, less than 5 stays.

3(iv). Round 12.292 to the nearest whole number, to tenths and to hundredths.

Whole number: 12. Tenths: 12.3. Hundredths: 12.29. Look at the next digit to the right: 5 or more rounds up, less than 5 stays.

3(v). Round 25.627 to the nearest whole number, to tenths and to hundredths.

Whole number: 26. Tenths: 25.6. Hundredths: 25.63. Look at the next digit to the right: 5 or more rounds up, less than 5 stays.

Exercise 6.4 - Add the following decimal numbers

1. Add 9.47 + 51.5.

Line up the decimal points and add column by column from the right, carrying where needed.

9.47 + 51.5 = 60.97.

Answer: 9.47 + 51.5 = 60.97.

2. Add 3.34 + 4.63 + 1.67.

Line up the decimal points and add column by column from the right, carrying where needed.

3.34 + 4.63 + 1.67 = 9.64.

Answer: 3.34 + 4.63 + 1.67 = 9.64.

3. Add 46.7 + 33.2.

Line up the decimal points and add column by column from the right, carrying where needed.

46.7 + 33.2 = 79.9.

Answer: 46.7 + 33.2 = 79.9.

4. Add 43.5 + 3.32.

Line up the decimal points and add column by column from the right, carrying where needed.

43.5 + 3.32 = 46.82.

Answer: 43.5 + 3.32 = 46.82.

5. Add 37.43 + 9.32.

Line up the decimal points and add column by column from the right, carrying where needed.

37.43 + 9.32 = 46.75.

Answer: 37.43 + 9.32 = 46.75.

6. Add 7.43 + 23.3 + 9.57.

Line up the decimal points and add column by column from the right, carrying where needed.

7.43 + 23.3 + 9.57 = 40.3.

Answer: 7.43 + 23.3 + 9.57 = 40.3.

Exercise 6.5 - Subtract the following decimal fractions

1. Subtract 7.93 from 8.23.

Line up the decimal points and subtract column by column from the right, borrowing where needed.

8.23 - 7.93 = 0.3.

Answer: 8.23 - 7.93 = 0.3.

2. Subtract 41.43 from 76.74.

Line up the decimal points and subtract column by column from the right, borrowing where needed.

76.74 - 41.43 = 35.31.

Answer: 76.74 - 41.43 = 35.31.

3. Subtract 6.54 from 9.78.

Line up the decimal points and subtract column by column from the right, borrowing where needed.

9.78 - 6.54 = 3.24.

Answer: 9.78 - 6.54 = 3.24.

4. Subtract 37.4 from 71.6.

Line up the decimal points and subtract column by column from the right, borrowing where needed.

71.6 - 37.4 = 34.2.

Answer: 71.6 - 37.4 = 34.2.

5. Subtract 7.84 from 9.98.

Line up the decimal points and subtract column by column from the right, borrowing where needed.

9.98 - 7.84 = 2.14.

Answer: 9.98 - 7.84 = 2.14.

6. Subtract 89.4 from 91.9.

Line up the decimal points and subtract column by column from the right, borrowing where needed.

91.9 - 89.4 = 2.5.

Answer: 91.9 - 89.4 = 2.5.

Exercise 6.6

1(i). Multiply 7.3 x 6.

Multiply as whole numbers: 73 x 6 = 438.

7.3 has one decimal place, so place the decimal point one place from the right in the product.

Answer: 7.3 x 6 = 43.8.

1(ii). Multiply 4.7 x 17.

Multiply as whole numbers: 47 x 17 = 799.

4.7 has one decimal place, so place the decimal point one place from the right in the product.

Answer: 4.7 x 17 = 79.9.

1(iii). Multiply 6.2 x 9.

Multiply as whole numbers: 62 x 9 = 558.

6.2 has one decimal place, so place the decimal point one place from the right in the product.

Answer: 6.2 x 9 = 55.8.

1(iv). Multiply 6.8 x 5.

Multiply as whole numbers: 68 x 5 = 340.

6.8 has one decimal place, so place the decimal point one place from the right in the product.

Answer: 6.8 x 5 = 34.0.

1(v). Multiply 8.4 x 12.

Multiply as whole numbers: 84 x 12 = 1008.

8.4 has one decimal place, so place the decimal point one place from the right in the product.

Answer: 8.4 x 12 = 100.8.

1(vi). Multiply 3.4 x 94.

Multiply as whole numbers: 34 x 94 = 3196.

3.4 has one decimal place, so place the decimal point one place from the right in the product.

Answer: 3.4 x 94 = 319.6.

2(i). Divide 6.4 by 2.

Divide as whole numbers and keep the same number of decimal places as the dividend.

6.4 / 2 = 3.2.

Answer: 6.4 / 2 = 3.2.

2(ii). Divide 7.5 by 5.

Divide as whole numbers and keep the same number of decimal places as the dividend.

7.5 / 5 = 1.5.

Answer: 7.5 / 5 = 1.5.

2(iii). Divide 8.4 by 6.

Divide as whole numbers and keep the same number of decimal places as the dividend.

8.4 / 6 = 1.4.

Answer: 8.4 / 6 = 1.4.

2(iv). Divide 9.1 by 7.

Divide as whole numbers and keep the same number of decimal places as the dividend.

9.1 / 7 = 1.3.

Answer: 9.1 / 7 = 1.3.

Exercise 6.7

1. A water tank can hold 5.5 litres. If 2.2 litres are already filled, how much more can be added?

More that can be added = capacity - already filled.

5.5 - 2.2 = 3.3.

Answer: 5.5 - 2.2 = 3.3 litres. Another 3.3 litres can be added.

2. A room is 4.5 metres long. If Aqila adds a 2.8 metres extension, what is the new length?

New length = old length + extension.

4.5 + 2.8 = 7.3.

Answer: 4.5 + 2.8 = 7.3 metres. The new length is 7.3 metres.

3. A bag contains 7.5 kg of wheat. If Imran buys 5 bags, how much wheat does Imran have?

Total wheat = weight of one bag x number of bags.

7.5 x 5 = 37.5.

Answer: 7.5 x 5 = 37.5 kg. Imran has 37.5 kg of wheat.

4. A 7.5 metres long rope is to be divided into 3 equal parts. What will be the length of each part?

Length of each part = total length / number of parts.

7.5 / 3 = 2.5.

Answer: 7.5 / 3 = 2.5 metres. Each part is 2.5 metres long.

Exercise 6.8 - Solve the following (DMAS: division, then multiplication, then addition, then subtraction)

1. 4 / 2 x 6 + 3 - 8.

First division: 4 / 2 = 2.

Then multiplication: 2 x 6 = 12.

Then addition: 12 + 3 = 15.

Then subtraction: 15 - 8 = 7.

Answer: 4 / 2 x 6 + 3 - 8 = 7.

2. 15 - 20 / 5 x 3 + 11.

First division: 20 / 5 = 4.

Then multiplication: 4 x 3 = 12.

Then work left to right: 15 - 12 = 3, then 3 + 11 = 14.

Answer: 15 - 20 / 5 x 3 + 11 = 14.

3. 15.3 / 3 + 3.6 - 0.5.

First division: 15.3 / 3 = 5.1.

Then addition: 5.1 + 3.6 = 8.7.

Then subtraction: 8.7 - 0.5 = 8.2.

Answer: 15.3 / 3 + 3.6 - 0.5 = 8.2.

4. 25.6 - 4.8 / 12 + 3.4.

First division: 4.8 / 12 = 0.4.

Then work left to right: 25.6 - 0.4 = 25.2, then 25.2 + 3.4 = 28.6.

Answer: 25.6 - 4.8 / 12 + 3.4 = 28.6.

5. 5/6 - 1/3 x 3/4.

First multiplication: 1/3 x 3/4 = 3/12 = 1/4.

Then subtraction: 5/6 - 1/4 = 10/12 - 3/12 = 7/12.

Answer: 5/6 - 1/3 x 3/4 = 7/12.

6. 7/8 / 7/16 + 1/2.

First division: 7/8 / 7/16 = 7/8 x 16/7 = 2.

Then addition: 2 + 1/2 = 2 1/2.

Answer: 7/8 / 7/16 + 1/2 = 2 1/2.

Review Exercise 6

1(i). Convert 7/10 to a decimal.

Change 7/10 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 7/10 = 0.7.

1(ii). Convert 45/20 to a decimal.

Change 45/20 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 45/20 = 2.25.

1(iii). Convert 913/200 to a decimal.

Change 913/200 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 913/200 = 4.565.

1(iv). Convert 447/100 to a decimal.

Change 447/100 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 447/100 = 4.47.

1(v). Convert 715/1000 to a decimal.

Change 715/1000 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 715/1000 = 0.715.

1(vi). Convert 69/50 to a decimal.

Change 69/50 into an equivalent fraction with denominator 10, 100 or 1000, then place the decimal point. 69/50 = 1.38.

2(i). Convert 0.79 to a fraction.

Write the digits after the decimal point over the matching place value and simplify. 0.79 = 79/100.

2(ii). Convert 75.7 to a fraction.

Write the digits after the decimal point over the matching place value and simplify. 75.7 = 757/10.

2(iii). Convert 0.010 to a fraction.

Write the digits after the decimal point over the matching place value and simplify. 0.010 = 1/100.

2(iv). Convert 0.005 to a fraction.

Write the digits after the decimal point over the matching place value and simplify. 0.005 = 1/200.

2(v). Convert 4.08 to a fraction.

Write the digits after the decimal point over the matching place value and simplify. 4.08 = 102/25.

2(vi). Convert 87.1 to a fraction.

Write the digits after the decimal point over the matching place value and simplify. 87.1 = 871/10.

3(i). Compare and order: 3.67, 3.59, 4.38, 5.67.

Ascending order: 3.59, 3.67, 4.38, 5.67. Descending order: 5.67, 4.38, 3.67, 3.59.

3(ii). Compare and order: 6.27, 6.54, 6.53, 6.21.

Ascending order: 6.21, 6.27, 6.53, 6.54. Descending order: 6.54, 6.53, 6.27, 6.21.

3(iii). Compare and order: 9.15, 9.1, 9.13, 9.19.

Ascending order: 9.1, 9.13, 9.15, 9.19. Descending order: 9.19, 9.15, 9.13, 9.1.

4(i). Add 6.21 + 2.1.

Line up the decimal points and add column by column from the right.

Answer: 6.21 + 2.1 = 8.31.

4(ii). Add 2.66 + 1.59.

Line up the decimal points and add column by column from the right.

Answer: 2.66 + 1.59 = 4.25.

4(iii). Add 7.34 + 1.21.

Line up the decimal points and add column by column from the right.

Answer: 7.34 + 1.21 = 8.55.

4(iv). Add 13.9 + 7.67.

Line up the decimal points and add column by column from the right.

Answer: 13.9 + 7.67 = 21.57.

4(v). Add 9.62 + 2.57.

Line up the decimal points and add column by column from the right.

Answer: 9.62 + 2.57 = 12.19.

4(vi). Add 7.14 + 0.25.

Line up the decimal points and add column by column from the right.

Answer: 7.14 + 0.25 = 7.39.

5(i). Subtract 5.78 from 24.6.

Line up the decimal points and subtract column by column from the right.

Answer: 24.6 - 5.78 = 18.82.

5(ii). Subtract 43.78 from 46.12.

Line up the decimal points and subtract column by column from the right.

Answer: 46.12 - 43.78 = 2.34.

5(iii). Subtract 12.75 from 56.89.

Line up the decimal points and subtract column by column from the right.

Answer: 56.89 - 12.75 = 44.14.

5(iv). Subtract 34.2 from 67.9.

Line up the decimal points and subtract column by column from the right.

Answer: 67.9 - 34.2 = 33.7.

5(v). Subtract 8.2 from 9.76.

Line up the decimal points and subtract column by column from the right.

Answer: 9.76 - 8.2 = 1.56.

5(vi). Subtract 23.5 from 63.9.

Line up the decimal points and subtract column by column from the right.

Answer: 63.9 - 23.5 = 40.4.

6(i). Multiply 8.4 x 6.

Multiply as whole numbers, then place the decimal point to match the decimal places in the question.

Answer: 8.4 x 6 = 50.4.

6(ii). Multiply 9.7 x 7.

Multiply as whole numbers, then place the decimal point to match the decimal places in the question.

Answer: 9.7 x 7 = 67.9.

6(iii). Multiply 3.4 x 9.

Multiply as whole numbers, then place the decimal point to match the decimal places in the question.

Answer: 3.4 x 9 = 30.6.

6(iv). Multiply 1.4 x 11.

Multiply as whole numbers, then place the decimal point to match the decimal places in the question.

Answer: 1.4 x 11 = 15.4.

6(v). Multiply 1.6 x 31.

Multiply as whole numbers, then place the decimal point to match the decimal places in the question.

Answer: 1.6 x 31 = 49.6.

6(vi). Multiply 3.8 x 18.

Multiply as whole numbers, then place the decimal point to match the decimal places in the question.

Answer: 3.8 x 18 = 68.4.

7(i). Divide 1.2 by 3.

Divide as whole numbers and keep the same number of decimal places as the dividend.

Answer: 1.2 / 3 = 0.4.

7(ii). Divide 6.5 by 5.

Divide as whole numbers and keep the same number of decimal places as the dividend.

Answer: 6.5 / 5 = 1.3.

7(iii). Divide 9.3 by 3.

Divide as whole numbers and keep the same number of decimal places as the dividend.

Answer: 9.3 / 3 = 3.1.

7(iv). Divide 9.9 by 33.

Divide as whole numbers and keep the same number of decimal places as the dividend.

Answer: 9.9 / 33 = 0.3.

7(v). Divide 6.3 by 21.

Divide as whole numbers and keep the same number of decimal places as the dividend.

Answer: 6.3 / 21 = 0.3.

7(vi). Divide 7.6 by 12.

7.6 / 12 does not divide exactly: 12 x 0.6 = 7.2, leaving 0.4, and the division repeats.

7.6 / 12 = 0.6333..., about 0.63 to two decimal places.

Answer: 7.6 / 12 = 0.6333... (approximately 0.63).

8. A container has 2.5 litres of juice. If Shazia adds 1.8 litres, what is the total quantity?

Total = 2.5 + 1.8 = 4.3.

Answer: 2.5 + 1.8 = 4.3 litres. The total quantity is 4.3 litres.

9. A bag of rice weighs 9.5 kg. If Majeed sells 3.5 kg, how much rice is left?

Left = 9.5 - 3.5 = 6.0.

Answer: 9.5 - 3.5 = 6.0 kg. There are 6.0 kg of rice left.

10. A bubble gum costs Rs. 2.5. If Wajid buys 3 bubble gums, how much will Wajid pay?

Cost = 2.5 x 3 = 7.5.

Answer: Rs. 2.5 x 3 = Rs. 7.5. Wajid will pay Rs. 7.5.

11. A 7.2 metre long wire is to be cut into 4 equal pieces. What will be the length of each piece?

Each piece = 7.2 / 4 = 1.8.

Answer: 7.2 / 4 = 1.8 metres. Each piece is 1.8 metres long.

Unit 7: Patterns

Unit 7: Patterns. Number patterns, rules, multiplication patterns, and simple expressions and number sentences.

Exercise 7.1

1(1). Complete the pattern: 2, 4, 6, 8, 10, ____, ____, ____, ____. State the rule.

Rule: add 2 each time (increasing by 2). The next four terms are 12, 14, 16, 18.

1(2). Complete the pattern: 17, 19, 21, 23, 25, ____, ____, ____, ____. State the rule.

Rule: add 2 each time (increasing by 2). The next four terms are 27, 29, 31, 33.

1(3). Complete the pattern: 48, 44, 40, 36, 32, ____, ____, ____, ____. State the rule.

Rule: subtract 4 each time (decreasing by 4). The next four terms are 28, 24, 20, 16.

1(4). Complete the pattern: 10, 15, 20, 25, 30, ____, ____, ____, ____. State the rule.

Rule: add 5 each time (increasing by 5). The next four terms are 35, 40, 45, 50.

1(5). Complete the pattern: 27, 25, 23, 21, 19, ____, ____, ____, ____.

State the rule.

Rule: subtract 2 each time (decreasing by 2). The next four terms are 17, 15, 13, 11.

1(6). Complete the pattern: 6, 12, 18, 24, 30, ____, ____, ____, ____. State the rule.

Rule: add 6 each time (increasing by 6). The next four terms are 36, 42, 48, 54.

1(7). Complete the pattern: 1, 3, 5, 7, 9, ____, ____, ____, ____. State the rule.

Rule: add 2 each time (increasing by 2). The next four terms are 11, 13, 15, 17.

1(8). Complete the pattern: 40, 38, 36, 34, 32, ____, ____, ____, ____. State the rule.

Rule: subtract 2 each time (decreasing by 2). The next four terms are 30, 28, 26, 24.

1(9). Complete the pattern: 12, 15, 18, 21, 24, ____, ____, ____, ____. State the rule.

Rule: add 3 each time (increasing by 3). The next four terms are 27, 30, 33, 36.

1(10). Complete the pattern: 32, 30, 28, 26, 24, ____, ____, ____, ____. State the rule.

Rule: subtract 2 each time (decreasing by 2). The next four terms are 22, 20, 18, 16.

2. Complete the multiplication table for 3, 6, 9, 12, 15 multiplied by 10, 100 and 1000.

Multiplied by 10: 30, 60, 90, 120, 150. Multiplied by 100: 300, 600, 900, 1,200, 1,500. Multiplied by 1000: 3,000, 6,000, 9,000, 12,000, 15,000. Multiplying by 10, 100 or 1000 only adds one, two or three zeros.

3. Find at least five patterns in the 100-chart and state their rules.

Five patterns: (1) Going down a column adds 10 each time. (2) Going right along a row adds 1 each time. (3) Every second number is even. (4) Multiples of 5 end in 0 or 5. (5) Multiples of 10 are the last column: 10, 20, 30, up to 100.

Exercise 7.2

1(i). Write an expression: a number x increased by 7.

x + 7.

1(ii). Write an expression: 5 less than a number x.

x - 5.

1(iii). Write an expression: 2 glasses of water a day for x days.

2x (2 multiplied by x).

1(iv). Write an expression: 60 shared equally among x children.

60 / x.

2(i). Write a number sentence and solve it: a number plus 7 equals

15.

Number sentence: x + 7 = 15. Subtract 7 from both sides: x = 15 - 7 = 8.

2(ii). Write a number sentence and solve it: a number minus 5 equals 8.

Number sentence: x - 5 = 8. Add 5 to both sides: x = 8 + 5 = 13.

2(iii). Write a number sentence and solve it: 3 times a number equals 12.

Number sentence: 3x = 12. Divide both sides by 3: x = 12 / 3 = 4.

2(iv). Write a number sentence and solve it: a number divided by 5 equals 4.

Number sentence: x / 5 = 4. Multiply both sides by 5: x = 4 x 5 = 20.

3(i). Find the rule: 4, 8, 12, 16, ...

Rule: add 4 each time. These are the multiples of 4.

3(ii). Find the rule: 6, 10, 14, 18, 20, ...

FLAG: the printed pattern breaks its own rule at the last term. The rule is add 4 each time, so after 18 the next term should be 22, not 20. The printed 20 is a book misprint; the rule is +4.

3(iii). Find the rule: 6, 12, 24, 48, ...

Rule: multiply by 2 each time (double the previous term).

3(iv). Find the rule: 41, 35, 29, 23, 17, 11, 5, ...

Rule: subtract 6 each time (decreasing by 6).

4(1). For y = x + 5, find y when x = 1, 2, 3 (except part (v), where even x values are used).

Pairs (x, y): (1, 6), (2, 7), (3, 8).

4(2). For y = 3 - x, find y when x = 1, 2, 3 (except part (v), where even x values are used).

Pairs (x, y): (1, 2), (2, 1), (3, 0).

4(3). For y = 3x + 1, find y when x = 1, 2, 3 (except part (v), where even x values are used).

Pairs (x, y): (1, 4), (2, 7), (3, 10).

4(4). For y = x squared, find y when x = 1, 2, 3 (except part (v), where even x values are used).

Pairs (x, y): (1, 1), (2, 4), (3, 9).

4(5). For y = x / 2, find y when x = 1, 2, 3 (except part (v), where even x values are used).

Pairs (x, y): (2, 1), (4, 2), (6, 3). Note: even values of x are used so that y is a whole number; x = 1 would give y = 1/2..

4(6). For y = 4x - 3, find y when x = 1, 2, 3 (except part (v), where even x values are used).

Pairs (x, y): (1, 1), (2, 5), (3, 9).

Review Exercise 7

1(1). Complete the pattern: 12, 14, 16, ____, ____, ____.

Rule: add 2 each time. The next three terms are 18, 20, 22.

1(2). Complete the pattern: 27, 29, 31, ____, ____, ____.

Rule: add 2 each time. The next three terms are 33, 35, 37.

1(3). Complete the pattern: 28, 24, 20, ____, ____, ____.

Rule: subtract 4 each time. The next three terms are 16, 12, 8.

1(4). Complete the pattern: 35, 40, 45, ____, ____, ____.

Rule: add 5 each time. The next three terms are 50, 55, 60.

1(5). Complete the pattern: 17, 15, 13, ____, ____, ____.

Rule: subtract 2 each time. The next three terms are 11, 9, 7.

1(6). Complete the pattern: 36, 42, 48, ____, ____, ____.

Rule: add 6 each time. The next three terms are 54, 60, 66.

1(7). Complete the pattern: 11, 13, 15, ____, ____, ____.

Rule: add 2 each time. The next three terms are 17, 19, 21.

1(8). Complete the pattern: 30, 28, 26, ____, ____, ____.

Rule: subtract 2 each time. The next three terms are 24, 22, 20.

1(9). Complete the pattern: 27, 30, 33, ____, ____, ____.

Rule: add 3 each time. The next three terms are 36, 39, 42.

1(10). Complete the pattern: 22, 20, 18, ____, ____, ____.

Rule: subtract 2 each time. The next three terms are 16, 14, 12.

2(a). A plant is 4 cm tall in week 1, 8 cm in week 2, 12 cm in week 3, 16 cm in week 4 and 20 cm in week 5. Find the pattern and the rule.

The height increases by 4 cm each week. Rule: height = 4 x number of weeks.

2(b). Blocks are stacked 15 in week 1, 35 in week 2, 55 in week 3, 75 in week 4, 95 in week 5 and 115 in week 6. Find the pattern and the rule.

The number of blocks increases by 20 each week. Rule: blocks = 20 x number of weeks - 5. Check: week 1 gives 20 - 5 = 15, week 6 gives 120 - 5 = 115.

3. Complete the multiplication table for 3, 12, 21, 30, 39 multiplied by 10, 100 and 1000.

Multiplied by 10: 30, 120, 210, 300, 390. Multiplied by 100: 300, 1,200, 2,100, 3,000, 3,900. Multiplied by 1000: 3,000, 12,000, 21,000, 30,000, 39,000.

4(1). For y = 5 - x, find y when x = 1, 2, 3.

Pairs (x, y): (1, 4), (2, 3), (3, 2).

4(2). For y = x + 4, find y when x = 1, 2, 3.

Pairs (x, y): (1, 5), (2, 6), (3, 7).

4(3). For y = 5x, find y when x = 1, 2, 3.

Pairs (x, y): (1, 5), (2, 10), (3, 15).

4(4). For y = 2x - 1, find y when x = 1, 2, 3.

Pairs (x, y): (1, 1), (2, 3), (3, 5).

Unit 8: Length, Mass and Capacity

Unit 8: Length, Mass and Capacity. Conversions: 1 km = 1,000 m, 1 m = 100 cm, 1 cm = 10 mm, 1 kg = 1,000 g, 1 litre = 1,000 millilitres.

Exercise 8.1

1. Convert kilometres to metres: 73 km, 14 km, 97 km, 87 km.

Multiply by 1,000. 73 km = 73,000 m; 14 km = 14,000 m; 97 km = 97,000 m; 87 km = 87,000 m.

2. Convert metres to centimetres: 3 m, 18 m, 27 m, 180 m.

Multiply by 100. 3 m = 300 cm; 18 m = 1,800 cm; 27 m = 2,700 cm; 180 m = 18,000 cm.

3. Convert kilometres to metres and then to centimetres: 3 km, 6 km, 11 km, 15 km.

3 km = 3,000 m = 300,000 cm; 6 km = 6,000 m = 600,000 cm; 11 km = 11,000 m = 1,100,000 cm; 15 km = 15,000 m = 1,500,000 cm.

4. Convert centimetres to millimetres: 31 cm, 90 cm, 110 cm, 125 cm.

Multiply by 10. 31 cm = 310 mm; 90 cm = 900 mm; 110 cm = 1,100 mm; 125 cm = 1,250 mm.

5. Convert 300 mm to centimetres.

1 cm = 10 mm, so divide by 10: 300 / 10 = 30.

Answer: 300 mm = 30 cm.

6. A ribbon is 1,500 mm long. 475 mm is cut off. How much is left?

Left = 1,500 - 475 = 1,025.

Answer: 1,500 - 475 = 1,025 mm are left.

7. Ali travels 128 km and then 142 km. What is the total distance?

Total = 128 + 142 = 270.

Answer: 128 + 142 = 270 km in total.

8. Convert 50 m to centimetres.

Multiply by 100: 50 x 100 = 5,000.

Answer: 50 m = 5,000 cm.

9. Add 1 km 450 m and 2 km 325 m.

Add kilometres: 1 + 2 = 3 km.

Add metres: 450 + 325 = 775 m.

Answer: 1 km 450 m + 2 km 325 m = 3 km 775 m.

10. Subtract 7 m 8 cm from 9 m 45 cm.

Subtract metres: 9 - 7 = 2 m.

Subtract centimetres: 45 - 8 = 37 cm.

Answer: 9 m 45 cm - 7 m 8 cm = 2 m 37 cm.

Exercise 8.2

1. Convert kilograms to grams: 95 kg, 71 kg, 67 kg, 55 kg, 9 kg, 11 kg, 17 kg, 29 kg.

Multiply by 1,000. 95 kg = 95,000 g; 71 kg = 71,000 g; 67 kg = 67,000 g; 55 kg = 55,000 g; 9 kg = 9,000 g; 11 kg = 11,000 g; 17 kg = 17,000 g; 29 kg = 29,000 g.

2. Convert 5 kg to grams.

Multiply by 1,000: 5 x 1,000 = 5,000.

Answer: 5 kg = 5,000 g.

3. Add 4,000 g and 500 g.

4,000 + 500 = 4,500.

Answer: 4,000 g + 500 g = 4,500 g.

4. Add 16,450 g and 12,325 g.

16,450 + 12,325 = 28,775.

Answer: 16,450 g + 12,325 g = 28,775 g, which is 28 kg 775 g.

5. Subtract 9,375 g from 15,800 g.

15,800 - 9,375 = 6,425.

Answer: 15,800 g - 9,375 g = 6,425 g, which is 6 kg 425 g.

6. A sack holds 50,000 g. 18,750 g are taken out. How much is left?

Left = 50,000 - 18,750 = 31,250.

Answer: 50,000 g - 18,750 g = 31,250 g are left.

7. Add 3,500 g and 4,250 g.

3,500 + 4,250 = 7,750.

Answer: 3,500 g + 4,250 g = 7,750 g.

Exercise 8.3

1. Convert litres to millilitres: 9 L, 74 L, 56 L, 67 L.

Multiply by 1,000. 9 L = 9,000 mL; 74 L = 74,000 mL; 56 L = 56,000 mL; 67 L = 67,000 mL.

2. Convert 15 L to millilitres.

Multiply by 1,000: 15 x 1,000 = 15,000.

Answer: 15 L = 15,000 mL.

3. Convert 13 L 50 mL to millilitres.

13 L = 13,000 mL, then add 50 mL: 13,000 + 50 = 13,050.

Answer: 13 L 50 mL = 13,050 mL.

4. Add 750 mL and 1 L 250 mL.

1 L 250 mL = 1,250 mL.

750 + 1,250 = 2,000 mL.

Answer: 750 mL + 1 L 250 mL = 2,000 mL, which is 2 L.

5. Subtract 20 L 750 mL from 25 L.

25 L = 25,000 mL and 20 L 750 mL = 20,750 mL.

25,000 - 20,750 = 4,250.

Answer: 25 L - 20 L 750 mL = 4,250 mL, which is 4 L 250 mL.

6. A tank held 425 litres. Now 39 litres are left. How many litres were used?

Used = total - left = 425 - 39 = 386.

Answer: 425 - 39 = 386 litres were used.

Review Exercise 8

1. Convert kilometres to metres: 5 km, 6 km, 1 km, 2 km, 16 km, 9 km.

Multiply by 1,000: 5,000 m; 6,000 m; 1,000 m; 2,000 m; 16,000 m; 9,000 m.

2. Convert metres to centimetres: 24 m, 218 m, 318 m, 532 m, 298

m, 128 m.

Multiply by 100: 2,400 cm; 21,800 cm; 31,800 cm; 53,200 cm; 29,800 cm; 12,800 cm.

3. Convert kilograms to grams: 68 kg, 321 kg, 41 kg, 12 kg, 25 kg, 112 kg.

Multiply by 1,000: 68,000 g; 321,000 g; 41,000 g; 12,000 g; 25,000 g; 112,000 g.

4. Convert metres to centimetres and then to millimetres: 5 m, 10 m, 13 m.

NOTE: the book prints 'millilitre (mm)' here; mm means millimetre.

5 m = 500 cm = 5,000 mm; 10 m = 1,000 cm = 10,000 mm; 13 m = 1,300 cm = 13,000 mm.

5. Add 23 km and 19 km.

23 + 19 = 42.

Answer: 23 km + 19 km = 42 km.

6. Subtract 1,000 m from 3,180 m.

3,180 - 1,000 = 2,180.

Answer: 3,180 m - 1,000 m = 2,180 m.

7. Add 49 kg and 65 kg, and give the answer in grams.

49 + 65 = 114 kg.

114 x 1,000 = 114,000 g.

Answer: 49 kg + 65 kg = 114 kg = 114,000 g.

8. Subtract 20 g 14 mg from 29 g 20 mg.

29 g 20 mg = 29,020 mg and 20 g 14 mg = 20,014 mg.

29,020 - 20,014 = 9,006.

Answer: 29 g 20 mg - 20 g 14 mg = 9,006 mg.

9. Four 1 litre bottles of juice are bought. After drinking some juice, 650 mL are left. How much juice was used?

FLAG: the book prints 'bought for 1 litre bottles'. This is read as 'four 1 litre bottles', the only reading that gives a total.

Total bought = 4 x 1,000 mL = 4,000 mL.

Used = total - left = 4,000 - 650 = 3,350 mL.

Answer: Four 1 litre bottles hold 4,000 mL in total; 4,000 - 650 = 3,350 mL were used.

10. A shop had 5,650 litres of oil. After selling some oil, 2,109 litres are left. How many litres were sold?

Sold = total - left = 5,650 - 2,109 = 3,541.

Answer: 5,650 - 2,109 = 3,541 litres were sold.

11. Add 1 L 450 mL and 2 L 275 mL.

1 L 450 mL = 1,450 mL and 2 L 275 mL = 2,275 mL.

1,450 + 2,275 = 3,725.

Answer: 1 L 450 mL + 2 L 275 mL = 3,725 mL.

Unit 9: Time and Date

Unit 9: Time and Date. Reading clocks, 12-hour and 24-hour time, converting hours, minutes and seconds, years, months, weeks and days, and finding elapsed time.

Exercise 9.1

Read the time on each clock. The pink hand is the hour hand and the blue hand is the minute hand, as in the lesson examples.

(i) 7:15, which is 15 minutes past 7. (ii) 3:30, which is half past 3. (iii) 9:25, which is 25 minutes past 9. (iv) 11:15, which is 15 minutes past 11. (v) 8:10, which is 10 minutes past 8. (vi) 1:45, which is 45 minutes past 1. Note: the times are given in hours and minutes, the form used in the lesson examples; seconds on the red second hand are not read here.

Exercise 9.2

1. Write in 12-hour time: (i) 19 minutes past 4 in the morning; (ii) quarter past 4 in the evening; (iii) quarter to 10 at night; (iv) half past 6 in the morning.

(i) 4:19 a.m. (ii) 4:15 p.m. (iii) 9:45 p.m. (iv) 6:30 a.m.

2. Convert to 24-hour time: 2:30 p.m., 11:15 a.m., 11:40 p.m., 7:25 a.m., 5:30 a.m., 9:45 p.m., 6:00 p.m., 8:33 a.m., 6:25 a.m., 3:40 p.m.

2:30 p.m. = 14:30; 11:15 a.m. = 11:15; 11:40 p.m. = 23:40; 7:25 a.m. = 07:25; 5:30 a.m. = 05:30; 9:45 p.m. = 21:45; 6:00 p.m. =

18:00; 8:33 a.m. = 08:33; 6:25 a.m. = 06:25; 3:40 p.m. = 15:40.

Exercise 9.3

1. Convert hours to minutes: 5 h, 6 h, 4 h, 7 h, 2 h, 9 h.

Multiply by 60. 5 h = 300 minutes; 6 h = 360 minutes; 4 h = 240 minutes; 7 h = 420 minutes; 2 h = 120 minutes; 9 h = 540 minutes.

2. Convert minutes to seconds: 18 min, 24 min, 13 min, 45 min, 60 min, 30 min.

Multiply by 60. 18 min = 1,080 seconds; 24 min = 1,440 seconds; 13 min = 780 seconds; 45 min = 2,700 seconds; 60 min = 3,600 seconds; 30 min = 1,800 seconds.

3. Convert hours to seconds: 2 h, 4 h, 5 h, 7 h, 3 h.

Multiply by 3,600. 2 h = 7,200 seconds; 4 h = 14,400 seconds; 5 h = 18,000 seconds; 7 h = 25,200 seconds; 3 h = 10,800 seconds.

Exercise 9.4

1. Convert years to months: 3 years, 6 years, 9 years, 12 years, 20 years.

Multiply by 12. 3 years = 36 months; 6 years = 72 months; 9 years = 108 months; 12 years = 144 months; 20 years = 240 months.

2. Convert months to days (taking 1 month = 30 days, as in this unit): 9 months, 11 months, 13 months, 3 months, 18 months.

Multiply by 30. 9 months = 270 days; 11 months = 330 days; 13 months = 390 days; 3 months = 90 days; 18 months = 540 days.

3. Convert weeks to days: 32 weeks, 59 weeks, 12 weeks, 13 weeks.

Multiply by 7. 32 weeks = 224 days; 59 weeks = 413 days; 12 weeks = 84 days; 13 weeks = 91 days.

Exercise 9.5

1. Find the time from 2:30 p.m. to 8:15 p.m.

From 2:30 p.m. to 8:30 p.m. is 6 hours.

8:15 p.m. is 15 minutes earlier, so subtract 15 minutes: 6 hours - 15 minutes = 5 hours 45 minutes.

Answer: 2:30 p.m. to 8:15 p.m. = 5 hours 45 minutes.

2. Find the time from 2:40 p.m. to 3:25 p.m.

From 2:40 p.m. to 3:00 p.m. is 20 minutes.

From 3:00 p.m. to 3:25 p.m. is 25 minutes.

20 + 25 = 45 minutes.

Answer: 2:40 p.m. to 3:25 p.m. = 45 minutes.

3(a). Find the time from 7:30 a.m. to 9:00 a.m.

From 7:30 a.m. to 9:00 a.m. is 1 hour 30 minutes.

Answer: 7:30 a.m. to 9:00 a.m. = 1 hour 30 minutes.

3(b). Find the time from 11:15 a.m. to 11:55 a.m.

11:55 - 11:15 = 40 minutes.

Answer: 11:15 a.m. to 11:55 a.m. = 40 minutes.

4. It is 10:30 p.m. What will the time be after 1 hour 40 minutes?

10:30 p.m. + 1 hour = 11:30 p.m.

11:30 p.m. + 40 minutes = 12:10 a.m. (past midnight).

Answer: After 1 hour 40 minutes it will be 12:10 a.m.

5. Find the time from 5 p.m. to 8:30 p.m.

From 5 p.m. to 8 p.m. is 3 hours, plus 30 minutes = 3 hours 30 minutes.

Answer: 5 p.m. to 8:30 p.m. = 3 hours 30 minutes.

Exercise 9.6

1. Add 1 hour 20 minutes and 1 hour 10 minutes.

Add hours: 1 + 1 = 2 hours.

Add minutes: 20 + 10 = 30 minutes.

Answer: 1 hour 20 minutes + 1 hour 10 minutes = 2 hours 30 minutes.

2. A watch is 15 minutes fast and shows 3:25 p.m. What is the actual time?

A fast watch shows a later time than the real time, so subtract 15 minutes: 3:25 - 0:15 = 3:10.

Answer: The actual time is 3:10 p.m.

3. Add 40 minutes and 1 hour 10 minutes.

1 hour 10 minutes = 70 minutes.

40 + 70 = 110 minutes = 1 hour 50 minutes.

Answer: 40 minutes + 1 hour 10 minutes = 1 hour 50 minutes.

4. Find the time from 13:00 to 14:20.

14:20 - 13:00 = 1 hour 20 minutes.

Answer: 13:00 to 14:20 = 1 hour 20 minutes.

5. One route takes 8 hours 30 minutes and another takes 6 hours 15 minutes. How much longer is the first route?

8 hours 30 minutes - 6 hours 15 minutes: hours 8 - 6 = 2, minutes 30 - 15 = 15.

Answer: 8 hours 30 minutes - 6 hours 15 minutes = 2 hours 15 minutes longer.

Review Exercise 9

1. Convert to 24-hour time: 7:45 p.m., 9:25 p.m., 10:40 p.m., 11:50 p.m.
7:45 p.m. = 19:45; 9:25 p.m. = 21:25; 10:40 p.m. = 22:40; 11:50 p.m. = 23:50.

2. Convert to 12-hour time: 16:15, 21:30, 18:25, 22:45.
16:15 = 4:15 p.m.; 21:30 = 9:30 p.m.; 18:25 = 6:25 p.m.; 22:45 = 10:45 p.m. FLAG: the printed key says '10:45 a.m.' for 22:45, which is wrong; 22:45 is at night, so it is p.m.

3. Convert hours to minutes: 1 h, 2 h, 3 h, 4 h.
Multiply by 60: 60, 120, 180, 240 minutes.

4. Convert to seconds: 3 minutes 10 seconds, 2 minutes 35 seconds, 4 minutes 15 seconds, 5 minutes 10 seconds.
3 minutes 10 seconds = 180 + 10 = 190 seconds. FLAG: the printed key says 100 seconds, which is wrong. 2 minutes 35 seconds = 120 + 35 = 155 seconds; 4 minutes 15 seconds = 240 + 15 = 255 seconds; 5 minutes 10 seconds = 300 + 10 = 310 seconds.

5. Convert years to months: 4 years, 3 years, 14 years, 21 years.
Multiply by 12: 48, 36, 168, 252 months.

6. Convert months to days (1 month = 30 days): 7 months, 5 months, 11 months, 1 month.
Multiply by 30: 210, 150, 330, 30 days.

7. A journey starts at the given time and takes three stages. Find the finishing time in each case.

(i) Start 4:30, then 20 minutes, then 45 minutes, then 70 minutes: 4:30 to 4:50 to 5:35 to 6:45. Finishing time 6:45. (ii) Start 2:10, then 35 minutes, then 25 minutes, then 75 minutes: 2:10 to 2:45 to 3:10 to 4:25. Finishing time 4:25. (iii) Start 11:45, then 25 minutes, then 105 minutes, then 75 minutes: 11:45 to 12:10 to 1:55 to 3:10. Finishing time 3:10. (iv) Start 6:20, then 45 minutes, then 12 minutes, then 65 minutes: 6:20 to 7:05 to 7:17 to 8:22. Finishing time 8:22. (v) Start 5:50, then 10 minutes, then 65 minutes, then 30 minutes: 5:50 to 6:00 to 7:05 to 7:35. Finishing time 7:35.

8. Add 11 hours 15 minutes and 1 hour 12 minutes.

Add hours: 11 + 1 = 12 hours.

Add minutes: 15 + 12 = 27 minutes.

Answer: 11 hours 15 minutes + 1 hour 12 minutes = 12 hours 27 minutes.

9. Subtract 4 hours 10 minutes from 6 hours 15 minutes.

Subtract hours: 6 - 4 = 2 hours.

Subtract minutes: 15 - 10 = 5 minutes.

Answer: 6 hours 15 minutes - 4 hours 10 minutes = 2 hours 5 minutes.

10. A shop opens at 9 a.m. and closes at 5 p.m. For how many hours is it open?

From 9 a.m. to 5 p.m. is 8 hours.

Answer: The shop is open for 8 hours.

Unit 10: Perimeter and Area

Unit 10: Perimeter and Area. Perimeter is the distance around a shape. Area is the space inside a shape. For a rectangle: perimeter = 2 x (length + breadth), area = length x breadth. For a square: perimeter = 4 x side, area = side x side.

Exercise 10.1

1(i). A rectangle is 3 cm long and 2 cm wide. Find its perimeter and area.

Perimeter = 2 x (3 + 2) = 2 x 5 = 10 cm.

Area = 3 x 2 = 6 square cm.

Answer: Perimeter = 10 cm, Area = 6 cm squared.

1(ii). A square has side 4 cm. Find its perimeter and area.

Perimeter = 4 x 4 = 16 cm.

Area = 4 x 4 = 16 square cm.

Answer: Perimeter = 16 cm, Area = 16 cm squared.

1(iii). A square has side 7 cm. Find its perimeter and area.

Perimeter = 4 x 7 = 28 cm.

Area = 7 x 7 = 49 square cm.

Answer: Perimeter = 28 cm, Area = 49 cm squared.

1(iv). A rectangle is 5 cm long and 3 cm wide. Find its perimeter and area.

Perimeter = 2 x (5 + 3) = 2 x 8 = 16 cm.

Area = 5 x 3 = 15 square cm.

Answer: Perimeter = 16 cm, Area = 15 cm squared.

1(v). Find the perimeter and area of the rectilinear shape in the book (sides 3, 3, 3, 2, 2, 3, 2, 2 cm in order around the shape).

Perimeter: add all eight sides: 3 + 3 + 3 + 2 + 2 + 3 + 2 + 2 = 20 cm.

Area: split the shape into a 3 cm by 3 cm part (9 square cm) and a 2 cm by 3 cm part (6 square cm): 9 + 6 = 15 square cm.

Answer: Perimeter = 20 cm, Area = 15 cm squared.

2. A wall is 4 m long and 3 m wide. Find its perimeter and area.

Perimeter = 2 x (4 + 3) = 14 m.

Area = 4 x 3 = 12 square m.

Answer: Perimeter = 14 m, Area = 12 m squared.

3. A square ground has side 8 cm. Find its perimeter and area.

Perimeter = 4 x 8 = 32 cm.

Area = 8 x 8 = 64 square cm.

Answer: Perimeter = 32 cm, Area = 64 cm squared.

Review Exercise 10

1. Find the perimeter and area of each rectangle: (i) 15 cm by 7 cm; (ii) 18.5 cm by 13 cm; (iii) 124 cm by 44 cm; (iv) 21 cm by 3 cm; (v) 26 cm by 13 cm.

(i) Perimeter = 2 x (15 + 7) = 44 cm, Area = 15 x 7 = 105 cm squared. (ii) Perimeter = 2 x (18.5 + 13) = 63 cm, Area = 18.5 x 13 = 240.5 cm squared. (iii) Perimeter = 2 x (124 + 44) = 336 cm, Area = 124 x 44 = 5,456 cm squared. FLAG: the printed key says 5,956 cm squared, which is wrong; 124 x 44 = 5,456. (iv) Perimeter = 2 x (21 + 3) = 48 cm, Area = 21 x 3 = 63 cm squared. (v) Perimeter = 2 x (26 + 13) = 78 cm, Area = 26 x 13 = 338 cm squared.

2. Find the perimeter and area of squares with sides 40 cm, 82 cm and 20 cm.

Side 40 cm: Perimeter = 4 x 40 = 160 cm, Area = 40 x 40 = 1,600 cm squared. Side 82 cm: Perimeter = 4 x 82 = 328 cm, Area = 82 x 82 = 6,724 cm squared. Side 20 cm: Perimeter = 4 x 20 = 80 cm, Area = 20 x 20 = 400 cm squared.

3(i). Find the perimeter and area of the L-shaped figure (top 3 cm,

left side 5 cm, bottom 8 cm, right side 3 cm, middle step 5 cm, inner step 2 cm).

Perimeter: add all six sides: 3 + 2 + 5 + 3 + 8 + 5 = 26 cm.

Area: split into a 3 cm by 5 cm part (15 square cm) and a 5 cm by 3 cm part (15 square cm): 15 + 15 = 30 square cm.

Answer: Perimeter = 26 cm, Area = 30 cm squared.

3(ii). Find the perimeter and area of the plus-shaped figure (a centre 2 cm by 2 cm square with four 2 cm by 2 cm arms).

Area: centre 2 x 2 = 4 square cm, plus four arms each 2 x 2 = 4 square cm: 4 + 16 = 20 square cm.

Perimeter: each arm adds 6 cm of outer edge, and there are four arms: 4 x 6 = 24 cm.

Answer: Perimeter = 24 cm, Area = 20 cm squared.

Unit 11: Practical Geometry

Unit 11: Practical Geometry. Parallel lines never meet and stay the same distance apart. A circle has a centre, a radius (centre to edge) and a diameter (edge to edge through the centre, twice the radius). Angles are measured in degrees with a protractor: less than 90 degrees is acute, exactly 90 degrees is a right angle, more than 90 degrees is obtuse.

Exercise 11.1

1. For each pair of lines, say whether they are parallel (mark with a cross) or not parallel (mark with a tick).

(i) Parallel, so mark with a cross. (ii) The lines cross, so they are not parallel: mark with a tick. (iii) The lines cross at a right angle, so they are not parallel: mark with a tick. (iv) Parallel vertical lines: mark with a cross. (v) The lines cross, so they are not parallel: mark with a tick. (vi) Parallel lines: mark with a cross.

Exercise 11.2

1. Name the centre, a radius and a diameter of each circle.

(i) Centre O, radius OA (or OB), diameter AB. (ii) Centre R, radius RP (or RQ), diameter PQ. (iii) Centre O, radius OM (or ON), diameter MN.

Exercise 11.3

1. Measure each angle with a protractor and say whether it is acute, right or obtuse.

(i) Angle MON = 90 degrees, a right angle (it carries the square corner mark). (ii) Angle GHI is about 105 degrees, an obtuse angle. (iii) Angle ABC is about 75 degrees, an acute angle. (iv) Angle JKL is about 135 degrees, an obtuse angle. (v) Angle TSU = 90 degrees, a right angle (it carries the square corner mark). (vi) Angle EFG is about 40 degrees, an acute angle. NOTE: the printed key gives no values for this exercise, so the four estimates are marked 'about' and should be checked with a protractor on the printed figure; the two right angles are exact because of their square corner marks.

2. Compare: 75 degrees and 30 degrees; 90 degrees and 120 degrees; 45 degrees and 45 degrees.

75 degrees is greater than 30 degrees. 90 degrees is less than 120 degrees. 45 degrees is equal to 45 degrees.

3. Draw angles with a protractor: (i) 30, 60, 45, 75 degrees; (ii) 90, 75, 105, 120 degrees; (iii) 155, 135, 105, 120, 90 degrees.

Method for each angle: draw a ray, place the centre of the protractor on the endpoint of the ray with the zero line along the ray, mark a point at the required number of degrees, remove the protractor and join the endpoint to the mark. The angles to draw are: (i) 30, 60, 45 and 75 degrees; (ii) 90, 75, 105 and 120 degrees; (iii) 155, 135, 105, 120 and 90 degrees.

Exercise 11.4

1. Name each angle, its vertex and its arms: (i) 70 degrees; (ii) 110 degrees; (iii) 90 degrees; (iv) 170 degrees.

(i) 70 degrees, vertex F, arms FC and FE. (ii) 110 degrees, vertex M,

arms MN and ML. (iii) 90 degrees, vertex R, arms RS and RQ. (iv) 170 degrees, vertex Y, arms YX and YZ.

2. Draw angles with a protractor: 130, 150, 20, 60, 40, 130, 80 and 105 degrees.

Use the protractor method from Exercise 11.3 Question 3: draw a ray, centre the protractor on its endpoint, mark the angle and join. FLAG: the printed key garbles two of these as 10 degrees and 90 degrees; the angles in the question itself (20 degrees and 130 degrees) are used here.

Exercise 11.5

1. Does the drawn line divide the shape into two mirror halves? (i) candy, diagonal line; (ii) hexagon, horizontal line; (iii) arrow, horizontal line; (iv) star, diagonal line.

(i) Candy, diagonal line: Yes. (ii) Hexagon, horizontal line: Yes. (iii) Arrow, horizontal line: No. (iv) Star, diagonal line: No.

2 and 3. Draw the mirror image of each shape across the line of symmetry, and state how many lines of symmetry each shape has.

Method: count the squares from each point of the shape to the symmetry line, then mark the mirror point the same number of squares on the other side, and join the points. Lines of symmetry: square 4, rectangle 2, circle infinite (any line through the centre), equilateral triangle 3, isosceles triangle 1, pentagon 5.

Review Exercise 11

1. Explain parallel and non-parallel lines, with examples.

Parallel lines never meet and stay the same distance apart, for example railway tracks, the opposite edges of a notebook, or the lines on a ruled page. Non-parallel lines meet, or would meet if extended, for example crossing roads, the hands of a clock, or a pair of scissors.

2. Name each angle, its size and its type.

(i) Angle ABC = 90 degrees, right angle. (ii) Angle KLM = 50 degrees, acute angle. (iii) Angle SNC = 100 degrees, obtuse angle (the key writes angle CNS, which is the same angle with vertex N). (iv) Angle PQR = 130 degrees, obtuse angle. (v) Angle HVW = 30 degrees, acute angle. (vi) Angle SUV = 170 degrees, obtuse angle. (vii) Angle DCE = 45 degrees, acute angle.

Unit 12: Data Handling and Probability

Unit 12: Data Handling and Probability. Bar graphs show data as bars: the scale and the labels must be read carefully. Probability words: impossible means it cannot happen, less likely means a smaller chance, equally likely means the same chance, more likely means a bigger chance, certain means it must happen.

Exercise 12.1

1. Chairs sold: 2008: 450; 2009: 370; 2010: 580; 2011: 620; 2012: 590. Draw a vertical bar graph and a horizontal bar graph, and interpret the data.

FLAG: the book lists the 2010 row twice (580 twice); it is a single 2010 entry and is counted once here. Most chairs were sold in 2011 (620) and fewest in 2009 (370). To draw the vertical bar graph: put the years on the horizontal axis and the number of chairs on the vertical axis with a scale of 1 unit = 100 chairs, then draw one bar per year to its height. For the horizontal bar graph: put the years on the vertical axis and the number of chairs on the horizontal axis with the same scale, then draw one bar per year to its length. The student draws the graphs from these values; this notes page gives the exact data table and the construction steps.

2. Marks table: Usman: English 91, Urdu 87, Maths 78, Science 98, Social Studies 92. Umar: English 97, Urdu 89, Maths 89, Science 88, Social Studies 87. Draw a double bar graph.

To draw the double bar graph: put the subjects on the horizontal

axis and the marks on the vertical axis, then draw two bars for each subject, one for Usman and one for Umar, using different colours and a key. The student draws the graph from this exact table; this notes page gives the data and the construction steps.

3. Oranges and apples sold (kg): Monday: oranges 15, apples 10; Tuesday: oranges 20, apples 10; Wednesday: oranges 25, apples 20; Thursday: oranges 15, apples 15; Friday: oranges 20, apples 15. Answer from the double bar graph.
(i) Most apples were sold on Wednesday (20 kg). (ii) Most oranges were sold on Wednesday (25 kg). (iii) Equal amounts were sold on Thursday (15 kg of each). (iv) Totals: oranges 15 + 20 + 25 + 15 + 20 = 95 kg; apples 10 + 10 + 20 + 15 + 15 = 70 kg. Oranges were sold in greater quantity.

Exercise 12.2

1. A spinner has 4 equal sections: red, blue, green and yellow. (i) Landing on red. (ii) Landing on purple. (iii) Landing on any colour.
(i) Red: 1 out of 4 sections, equally likely. (ii) Purple: there is no purple section, so it is impossible. (iii) Any colour: every section has a colour, so it is certain.

2. A spinner is numbered 1 to 6. (i) Landing on an odd number. (ii) Landing on a number greater than 6.
(i) Odd numbers are 1, 3 and 5: 3 out of 6 sections, equally likely.
(ii) No number is greater than 6, so it is impossible.

3. A spinner has 3 equal parts: red, blue and green. (i) Landing on

blue. (ii) Landing on yellow. (iii) If all parts were red, landing on red. (iv) If 2 parts were red and 1 part blue, landing on blue. (v) In that case, landing on red.

(i) Blue: 1 out of 3, equally likely. (ii) Yellow: there is no yellow part, so it is impossible. (iii) All parts red: landing on red is certain. (iv) Blue: 1 out of 3, less likely than red. (v) Red: 2 out of 3, more likely than blue.

4. Roll a die once. (i) Less than 4. (ii) Rolling 7. (iii) Getting a number from 1 to 6. (iv) Odd number. (v) Getting 3 compared with getting an odd number.

(i) Less than 4 means 1, 2 or 3: 3 out of 6, equally likely (exactly half). FLAG: the printed key calls this 'more likely', which is incorrect. (ii) Rolling 7: impossible. (iii) Getting 1 to 6: every face is 1 to 6, so it is certain. FLAG: the printed key calls this 'equally likely', which is incorrect for the whole set. (iv) Odd means 1, 3 or 5: 3 out of 6, equally likely. FLAG: the printed key calls this 'more likely', which is incorrect. (v) Getting 3 is 1 out of 6, while getting an odd number is 3 out of 6, so getting 3 is less likely.

Review Exercise 12

1. Read the bar graph of animal weights (scale: 1 unit = 2,000 kg). (a) Heaviest: White Shark, about 16,000 kg. (b) The White Shark (16,000 kg) is 4 times heavier than the Indian Elephant (4,000 kg): 16,000 / 4,000 = 4. (c) The Giraffe and the Crocodile have equal weights. (d) The animal weighing 2,800 kg is the White Rhinoceros. (e) Animals weighing more than 2,000 kg and less than 9,000 kg: African Elephant (8,000 kg), Indian Elephant (4,000 kg) and White

Rhinoceros (2,800 kg). FLAG: the printed key lists only the two elephants for part (e), omitting the rhinoceros although 2,800 kg is in range; all three are given here. (f) Increasing order: Crocodile, Giraffe, Hippo, White Rhinoceros, Indian Elephant, African Elephant, White Shark.

2. School enrolment (number of students): Girls: 2008: 16, 2009: 12, 2011: 18, 2012: 32, 2013: 36. Boys: 2008: 26, 2009: 32, 2011: 16, 2012: 28, 2013: 36. Draw a double bar graph.
Note: the year 2010 is absent from the book's table as printed, and is not invented here. To draw the graph: put the years on the horizontal axis and the number of students on the vertical axis, then draw two bars for each year, one for girls and one for boys, using different colours and a key. The student draws the graph from this exact table; this notes page gives the data and the construction steps.

3. Glasses of water drunk: Ali 16, Imran 10, Sohail 12, Faisal 8, Haris 10. Draw a horizontal bar graph and interpret it.
Total: 16 + 10 + 12 + 8 + 10 = 56 glasses. Most: Ali (16 glasses). Least: Faisal (8 glasses). To draw the horizontal bar graph: put the names on the vertical axis and the number of glasses on the horizontal axis with a scale of 1 large division = 2 glasses, then draw one bar per person to its length. The student draws the graph from these values; this notes page gives the exact data and the construction steps.

4. A spinner has 8 equal sections numbered 1 to 8 (there is no

yellow section). Find each probability.

(i) Yellow: 0 out of 8 = 0, impossible. (ii) Odd numbers (1, 3, 5, 7): 4 out of 8 = 1/2. (iii) Greater than 2 (3, 4, 5, 6, 7, 8): 6 out of 8 = 3/4. (iv) Less than 9 (all sections): 8 out of 8 = 1, certain.

5. Roll a die once. Find each probability.

(i) Even number (2, 4, 6): 3 out of 6 = 1/2. (ii) Multiple of 3 (3, 6): 2 out of 6 = 1/3. (iii) Number 5: 1 out of 6. (iv) Less than 6 (1, 2, 3, 4, 5): 5 out of 6. (v) Greater than 0 (all faces): 6 out of 6 = 1, certain.

6. What is the probability that a day of the week starts with the letter S?

Days starting with S are Saturday and Sunday: 2 out of 7 days. The probability is 2/7.

A note on flagged items in the printed book

This page records every place where these notes differ from the printed book, and why. Nothing was invented to fill a gap.

Coverage: what these notes cover.

These notes cover all numbered exercises and all review exercises in Units 1 to 12 of Mathematics-420, the complete book. In-text Activities are classroom tasks (group work, measuring and drawing in class), distinct from the numbered exercises; they are not written out as solved items, and the student does the activity in class. The Glossary (printed page 162) is a word list, not an exercise, so it has no solved items.

Printed-book errors corrected in these notes (the correct mathematics is given in the exercise itself).

Unit 1: the printed key says 13,076 is less than 10,772 (it is greater); worked Example 7 swaps the ascending and descending labels; the Roman table misprints XLII as 41 (correct 42); the words key gives 'Fifty thousand and one' for 50,006 (correct 'Fifty thousand and six'). Unit 5: the 'Remember!' box shows 7/3 as 2 2/8 with divisor 8 (correct 2 1/3, divisor 3). Unit 6: the worked examples swap 427/100 and 4321/1000 (correct 4.27 and 4.321); the key misprints 9.47 + 51.5 as 6.97 (correct 60.97) and 43.5 + 3.32 as 46.80 (correct 46.82). Review 4: the key marks 986 divisible by 3 (digit sum 23, so it is not). Review 9: the key gives 10:45 a.m. for 22:45 (correct 10:45 p.m.) and 100 seconds for 3 minutes 10

seconds (correct 190 seconds). Review 10: the key gives 5,956 cm squared for 124 x 44 (correct 5,456). Exercise 11.4 Question 2: the key garbles two angles as 10 degrees and 90 degrees (the question's 20 degrees and 130 degrees are used). Probability wording in the printed key is scrambled in places (Exercise 12.2 Question 4 and Review 12 Question 1 part (e), which omits the White Rhinoceros); the mathematically correct terms and the full list are given.

Printed-book misprints flagged, not silently fixed.

Exercise 5.5 Question 1 part (x): the first factor is missing in print (the box starts with a multiplication sign); only the printed part is worked (7 5/41 x 11 = 78 14/41) and nothing is guessed. Exercise 5.6 Question 4: the question prints '2' followed by a gap before 'pizzas'; it is solved as 2 pizzas (1/2 pizza each for 4 friends) and the gap is flagged. Exercise 7.2 Question 3 part (ii): the printed '20' breaks the +4 pattern (the next term should be 22). Exercise 12.1 Question 1: the 2010 row is printed twice (580 twice) and is counted once. Review 8 Question 4: the book prints 'millilitre (mm)' where millimetre is meant. Review 8 Question 9: the book prints 'for' where 'four' is meant. Exercise 6.7 Question 3: the book says 'rice' where wheat is meant.

Estimates, not exact key values.

Exercise 11.3 Question 1: the printed key gives no values at all for this exercise. Two angles are exactly 90 degrees (square corner marks). The other four measures (about 105, 75, 135 and 40 degrees) are visual estimates and are marked 'about'; check them

with a protractor on the printed figure.

Graphs and drawings.

Bar graphs (Exercise 12.1 Questions 1 and 2, Review 12 Questions 2 and 3) and protractor drawings and mirror images (Exercise 11.3 Question 3, Exercise 11.4 Question 2, Exercise 11.5 Questions 2 and 3) must be drawn by the student: this notes builder draws vector shapes only, not data bar charts or protractor figures. Each such item therefore gives the exact data, the construction steps and the interpretation, so the student can draw it correctly.